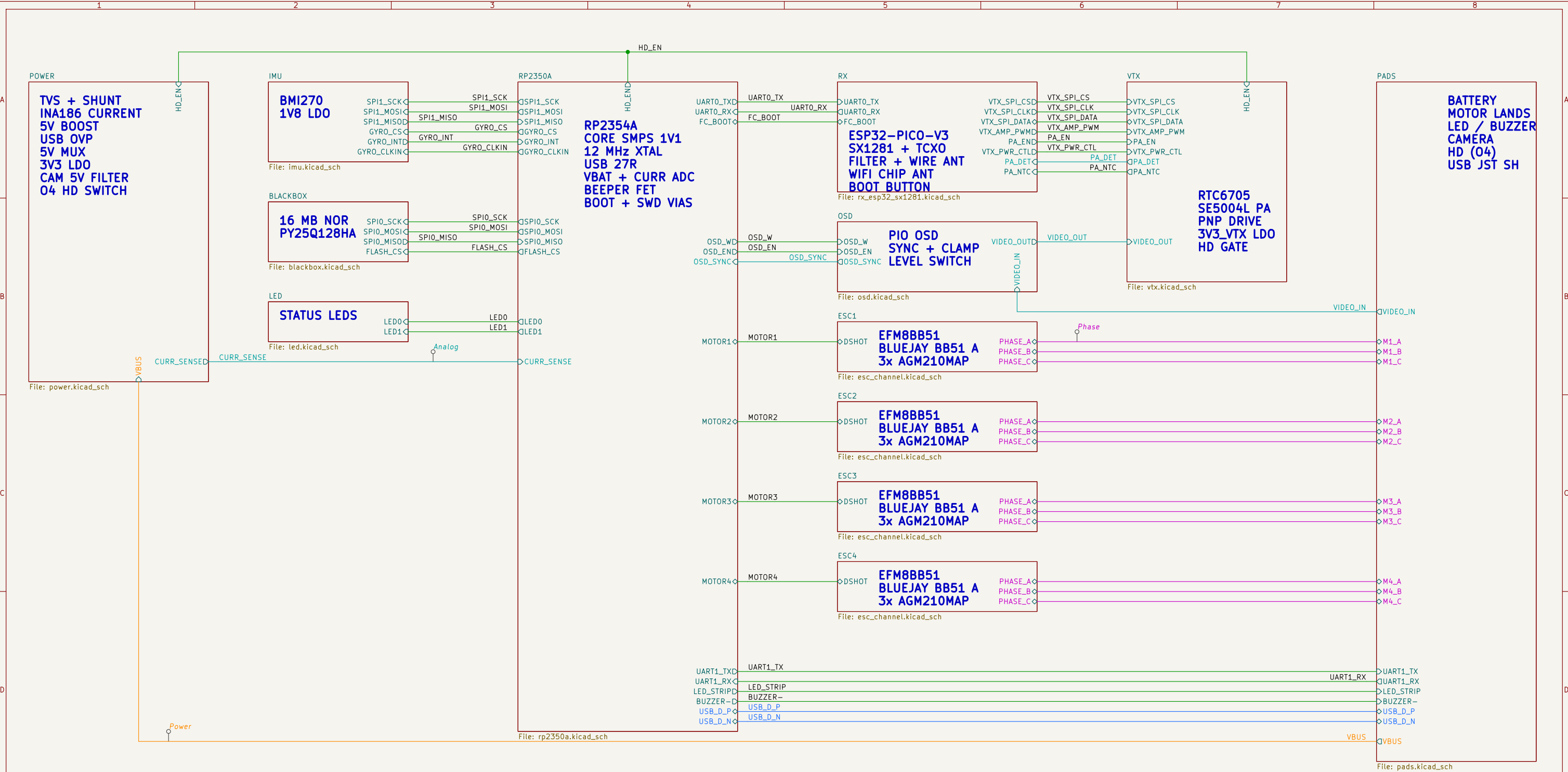




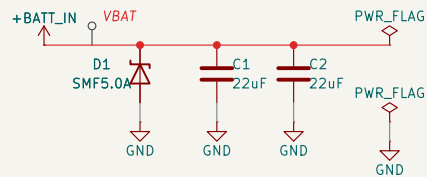
POWER TABLE

Rail	Source	Nominal	Budget (spec 5)	Feeds
+BATT_IN	battery pads J2 / J3 (1S)	3.0–4.35 V	4x ESC + 2.0 A board	TVS, pad bulk, shunt R1
+BATT	shunt R1 0.5 m0hm	cell	ESC + boost in 1.4–3.0 A	ESC1–4, boost, 04 EN, VBAT ADC, PA_VCC
/VBUS	USB J31 pin 4 (hier. net)	5 V	0.42 A, USB only	U24 OVP
+5V_USB	U24 SGM40661 OVP	5 V	as VBUS	U3 VIN1, PR1 divider
+5V_BST	U2 TPS61022 boost	5.15 V	0.70 A analog / 1.48 A HD (BEC 1.5 A cont.)	U3 VIN2, U5 04 switch
+5V	U3 TPS2116 mux	5 V	0.70 A analog, about 0.25 A HD	3V3 LDOs, CAM filter, 5V pads
+5V_CAM	FB1 ferrite from +5V	5 V	0.12 A	camera pad + plug
+5V_HD	U5 TPS22810 (HD_EN, cell)	5 V	1.20 A 04 Lite	VHD pad
+3V3	U4 LP5912–3.3	3.3 V	0.20 A (0.48 A Wi–Fi bench)	FC, IMU, NOR, OSD, RX, LEDs, INA186
+3V3_VTX	U21 LP5912–3.3 (off in HD)	3.3 V	0.10 A typ, about 0.17 A at max PA	RTC6705, PA drive, PA ref
+1V8	U12 TPS7A2018	1.8 V	< 1 mA	BMI270
+1V1	RP2354A core SMPS	1.1 V	internal	RP2354A DVDD
PA_VCC	R67 selector, +BATT default	cell	0.65 A at 27 dBm	SE5004L PA (local, VTX)
PA_VREF	U22 LP5907–2.85 (EN = PA_EN, off in HD)	2.85 V	about 10 mA	SE5004L VREF/EN (local, VTX)
GND	battery –	0 V	–	all sheets

Generated from hardware/sch_contract.json by hardware/tools/sch_root.py; do not edit by hand.
Root wires carry the project-wide net names (root label = net /NAME). Motor phases are named by the ESC sheets (/ESCn/PHASE_x). Rails are power symbols (table); VBUS is a hierarchical net, ESC1–ESC4 drive M1 rear right, M2 front right, M3 rear left, M4 front left (Betaflight quad–X); FC pin map: research/PINMAP.md.

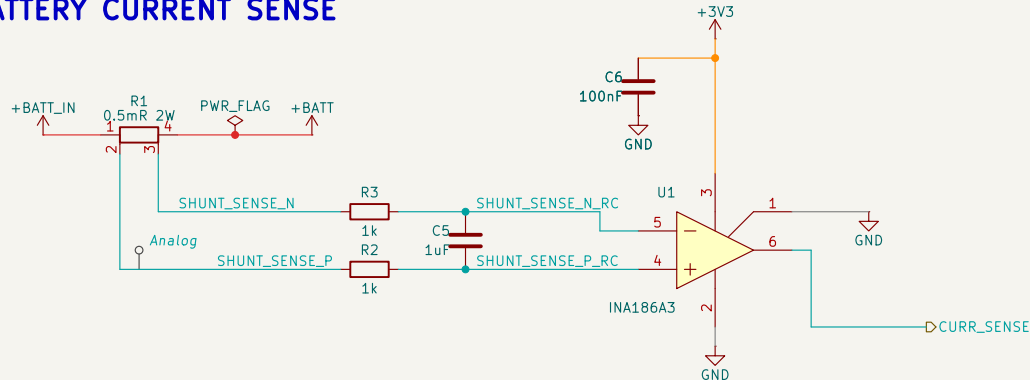
OpenDrone

BATTERY INPUT AND TVS



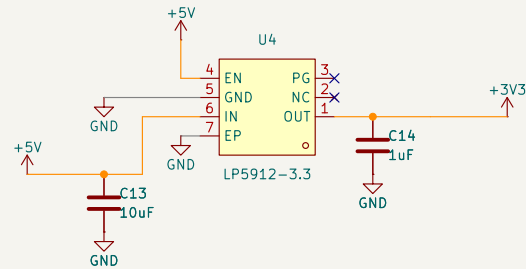
Source: OpenAIO/hardware/OpenAIO.kicad.sch (battery entry), adapted.
 +BATT1N = battery pads J2 (+) / J3 (-) on the PADS sheet, 1S LiPo/LiHV
 3.0–4.35 V; PWR_FLAG here (the pads are passive). No reverse–polarity FET (spec
 4.1: user–fitted pigtail, risk in README).
 D1 SMF5.0A: VBR 6.40–7.00 V (10 mA), VC 9.2 V at 21.7 A, 200 W 10/1000 us;
 takes one motor's regenerative energy (–0.29 J per rated pulse) on a battery disconnect;
 fallback SRAJ5.0A (SMA, 400 W) if V3b exceeds effect. It does not hold the 5.5–7 V
 parts in rating (EF86 VDD on the cell; TP561022 VIN 7 V abs max, and its
 pins through–hole; V3b VIN 7 V abs max, and its pins through–hole; +5V_BST =
 TP52116 VIN2, 6 V abs max);
 Matrix–parity risk spec 15.4.
 C1/C2 22 uF 16 V X5R 0603 at the pads: 9–13 uF effective each at 4.35 V. Hot
 plug of a 4.35 V pack can overshoot to 5.0–5.4 V (V3C): above the TP561022 4.8
 V VIN limit for a start–up with VOUT pre–biased –0.5 V (7C). (DS 6.3), which is
 exactly hot plug (EN = VIN, +5V_BST at 0 V); V3C scopes VIN with the +5V_BST
 start. Damping fallback –100 uF polymer at the pads: PE–8 waits for V3C.

BATTERY CURRENT SENSE



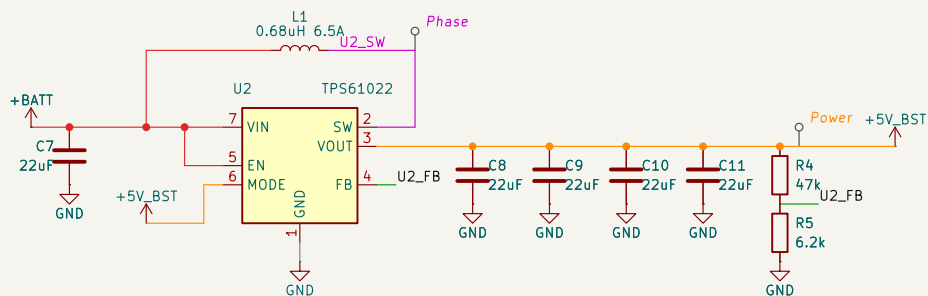
Source: OpenAIO/hardware/OpenAIO_1A00_sch (shunt + INA186A3, root sheet); input RC moved to 0201 and its two common-mode caps deleted (DS2 BR-07); the output RC (1k/100n) sits at the ADC pin (RC2350A sheet, R47/C60).
P4-9: R1 HCS1206 0.5 mOhm on the four-pad Kelvin net-tie land (pad groups 1-2 and 3-4). SHUNT_SENSE_PC = pad 2 (battery side).
SHUNT_SENSE_N = pad 3: each holds only its inner sense pad and R2/R3; SHUNT_SENSE_PC/N_RC hold R2/R3, C5 and U1 (netlist checked); all four match the Analog and Kelvin DRL patterns. Route the pair on L1 or L6, never ln2/ln3 (DRU), class Analog. Everything that draws current taps the sense line.
R2/R3 R1 + C5 1 uF differential: $f_c = 1/(2 \pi \times 2 \times 0.5 \text{ m}\Omega \times 1 \text{ }\mu\text{F}) = 80 \text{ Hz}$, $\text{ILB} < 3 \times \text{INA} \times 1 \text{ k}\Omega \times 3 \text{ }\mu\text{V} (6 \text{ mA})$, $V_{OS} < 50 \text{ }\mu\text{V} (0.1 \text{ A})$.
U1 INA186A3, gain 100: 0.5 mOhm to 50 mV/A. Betaflight ibata-scale 500; output to VS - 40 mV gives about 65 A full scale on +3V3; VCM -0.2..0.4 V covers the cell; REF to GND (unidirectional); C6 100 nF at VS. Shunt loss 0.2 W at 20 A. Scale checked on every prototype (V-plan).

3.3 V LDO FOR FC + RX



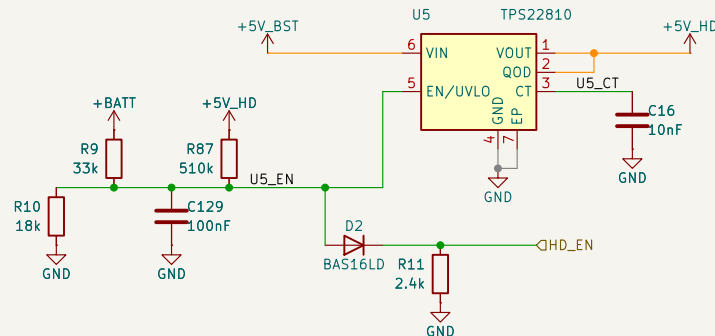
Source: OpenFC-Lite-Mini/hardware/power.kicad_sch (3.3V 0.5A MAIN LDO); LP5912-3.3, with the OpenDrone LP5912-3.3DRVR symbol (pins in package order: IN/EN on the left here, EN tied to +5V, no EN shared with another rail; U21 on VTX has its own EN), PG and NC open; EP to GND for heat (TI: GND or open).
P4-17 / BR-14: C13 10 uF x6S 0402 is the one CIN of U4 and U21 (VTX sheet): place it between the two IN pins (-2.4 mm apart). 2.0 uF effective at 5 V (Murata curve) > the 0.7 uF LP5912 minimum.
P4-12, +3V3 OUT at 3.3 V: C82 10 uF 3.5 uF (Murata GRM155C80J106ME11 DC-bias curve), C58 4.7 uF 0402 X5R -2.6 uF, 3x 1 uF 0201 X5R (C92, C93, C94) for each +3V3 uF x6S 0201 (6) for the rest of that chip type (0.1); 8.7 uF typical 7442 uF at -20, 10.3 uF only with every part at 0.0 uF -> inside TI's 0.7-10 uF (SNSVAT47-06) but for that +3 V corner, and most of it is remote (=50 mm of trace), which the LP5912 allows. Lower if a V-bench load step rings: PS-06 (delete) C92, +3V3_VTX is summed on the VTX sheet.
Load 0.20 A: (5.15 - 3.3) V x 0.2 A = 0.37 W, 0.26 K (RthJA 71.2 K/W). U4 OUT (project-lib LP5912 copy, power_out pin) drives +3V3: no PWR_FLAG.

5 V BOOST TPS61022



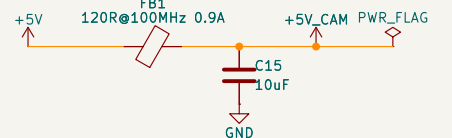
New block from TI: SLVSDXD7D (nrsilbing uses the TPS61022); look matched to the OpenFxC-Lite-Mini buck blocks.
P4-7: VOUT = 0.600 V $\times$ (1 + 47k / 6.2k) = 5.148 V, with VFB 585=615 mV and 1 $\times$ resistors 4.93=5.37 V. R5 6.2k < 300 kOhm (TI). EN tied to VIN (+BATT) and nothing else (BR-11); same -0.3..7 V abs max, VOUT_H <= 1.2 V, on whenever the cell is above UVLO. MODE = VOUT (+5V_BST); forced PWM, VMODE_H <= 1.2 V valid once VOUT > 2.2 V (start-up pass-through holds VOUT near VIN).
C7 22 uF CIN at VIN/GND. C8-C11 4x 22 uF X6S 0603; -20-28 uF effective at 5.15 V, inside TI's 10=50 uF; TI 6.3 asks >= 10 uF effective up to 1.5 A and >= 20 uF from 1.5 to 3 A, met only typically, so the published BEC is 1.5 A continuous (D79: HD load 1.43=1.48 A, LED strip loaded in HD mode; V1 load step at 1.7 A). Feed-forward cap (TI 8.2.2.4: above 40 uF effective, fFZ 2 kHz = 1.7 nF across R4); C8-C11 give 20=28 uF, HD mode adds C12/C13 (-6 uF via the mux) and the O4 input (-47 uF, 20 uF): none fitted (lean). V1b steps the load in HD with a real O4, 1.5=2.2 nF across R4 only if it rings. L1 0.68 uH, Isat 6.5 A, 17 mOhm; valley limit 6.5=10 A; converter capability 1.67/1.82/1.98 A at 2.8/3.0/3.2 V (spec 4.2).
Passes VIN to VOUT when VIN > VOUT (surge path, spec 4.1).
Layout (TI 10.1): Cout VOUT/GND pads at pins 3/1 (hot loop SW-VOUT-Cout-GND smallest). CIN at pins 7/1. SW copper minimal. R4/R5 at FB pin 4 away from SW; ground plane under the part.

04 LITE HD SUPPLY SWITCH TPS22810



New block from TI SLVSDHOC (D15, D34; QD53 BR-10/22/24), U5 = TPS22810DPRV WSON-6 2x2 (D83, P4 r4; smallest package, 3 A at TA 65 C vs 2 A DBV, RthJA 74.6 vs 182 C/W; same die and EN/UVLO figures: pins 1 VOUT2 D2 Q3 3 CT 4 GND/5 EN/UVLO 6 VIN, EP7 T0 to GND as on TI's EVM, SLVU478 Fig. 1). P4-2 / BR-10: U5 VIN on +5V_USB, ahead of the mux: USB never reaches the U5. No separate CIN (C17 deleted, P4 critique): the boost outputs C8-C11 (20-28 uF effective) are U5's input capacitor (TI 10.3: CIN optional, 1 uF usually sufficient); P5 condition: U5 VIN within 3 mm of C8-C11 or U3 VIN2 on the +5V_USB copper. C16 CT 10 nF: -4.7 V/mS, 0.2 A inrush per 47 uF. Q4 tied to VOUT; the internal RPD empties +5V_HD when off. U5_EN (bom_plan Q4Q4) holds exactly R9 33k / R10 18k from +BATT, R87 510k from +5V_HD, C129 100 nF, D2 anode and U5 EN/UVLO pin 5 (netlist corrected, D2 BAS16LD cathode + R11 2.4k pull-down on HD_EN (GPIO27, PINIO1)). P4-1 (VENV 1.3-1.50 V, VENV 1.08-1.18 V, +5V_HD 4.93-5.37 V): EN = 0.345 x cell (+0.0223 x +5V_HD when on). Nominal R: on at 3.27-3.77 V, shed at 2.78-3.10 V; 1 % R: on at 3.4-3.6 V; 1 % R: W: hysteresis 0.32-0.340 V => the 0.31 x cell rebound is not a problem. GPIO27 low or undriven: EN <= VF 0.715 + 66 uA x 2.4k = 0.87 V < VENV 1.08 V, also with +5V_HD still up. GPIO27 high: D2 reverse >= 1.7 V; 1 uA hot leakage moves EN 11 mV (shed 0.03 V); R11 draws 1.4 mA. Filter: 11.4 kOhm x 100 nF = 1.14 mS (140 Hz): the 96 kHz ESC ripple (0.21 V p-p at EN) falls to 0.3 mV, D2 + R11 discharge it below VENV in 0.25 mS; turn-on: from the clamp -1.6 mS at 4.2 V. USB only (back-fed 2.5-2.8 V + 0.3 V ripple): EN <= 1.07 V < VENV, EN <= 1.62 V <= 1.8 V abs max.

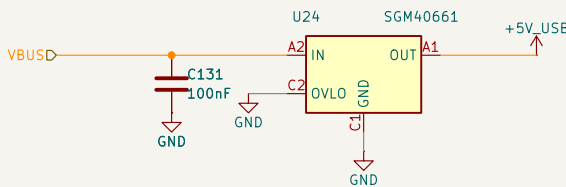
CAMERA 5 V FILTER



Source: OpenFC-Lite-Mini/hardware/power.kicad_sch (camera 5 V feed), with the commons 120 Ohm 0201 bead (blocks/LDO_IMU_TPS7A2033).

Fb1 BLM03P3X125N1D: 120 Ohm at 100 MHz, 0.9 A, 160 mOhm; the 0.12 A camera drops 19 mV; it keeps the boost's 1 MHz and the camera's own switching apart. C15 10 uF X6S after the bead, +5V_CAM feeds the CAM pad J24 and the camera plug J32 (PADS); PWR_FLAG because FB1 is passive. Falback BLM03P3X220S1N1 (1.45 A).

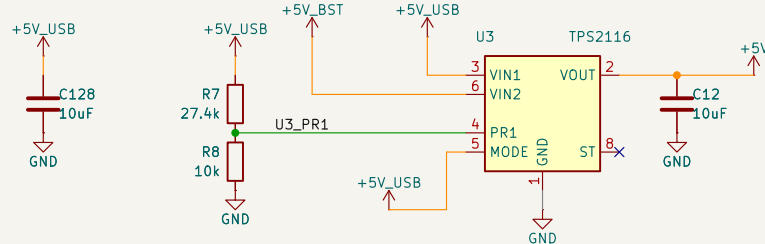
USB INLET OVP SGM40661



New block from the SGMICRO SGM40659/60/61 data sheet REV.A (D57).
 IN = VBUS (J31 pin 4, PADS). OUT = +5V_USB. OVLO (C2) to GND selects the internal threshold:
 5.75–6.12 V rising, 5.45–5.66 V falling, IN 28 V abs max, 2.7–20 V recommended (feature limit: 40 V surge immunity), OVULO 2.4 V, 15 ms debounce, then a 2 ms soft turn-on: a cable LC ring (simulated <= 10.8 V at IN) never reaches TPS2116 VIN1 (6 V abs max), and the soft turn-on is the VBUS surge limit USB asks for above 10 V downstream.
 C131 100 nF X65 25 V at the IN balls (data sheet CIN 0.1 uF; also takes ESD charge). COUT = 220 nF at U3 VIN1 (<= 100 uF allowed). RON 27 / 45 mOhm; 11–19 mV at 0.42 A. TA = 85 C.
 Open: a non-compliant source at 5.5–6.12 V would pass before the OVLO trips (TPS2116 VIN1 5.5 V recommended, 6 V abs max); USB sources are <= 5.25 V (5.5 V with PD).

Open: a non-compliant source at 5.5–6.12 V would pass before the OVLO trips (TPS2116 VIN1 5.5 V recommended, 6 V abs max); USB sources are ≤ 5.25 V (5.5 V with PD).

5 V POWER MUX TPS2116



New block from TI SLVSFG1A, priority mode (D24). C128 = the VIN1 cap; place it at U3 pin 3 (Ti 9-10).
 Mode tied to VIN1 (+5V_USB). PR1 = VIN1 x 10k / 374k; USB is selected above VREF x 3.74 = 3.44-4.04 V (VREF 0.92-1.08 V:
 3.37-4.10 V with 1 % R), +5V_BST (VIN2) otherwise. ST (open drain) unused.
 P4-2: +5V_BST holds only U2 VOUT/VMODE, C8-C11, R4, U3 VIN2 and U5 VIN (netlist checked). USB selected: VIN2 stays off (truth
 table "VIN2 = X") and both channels block reverse current; so USB never feeds +5V_BST or the O4.
 P4-3: C128 10 uF x65 0402 at VIN1 (TI CIN 1 uF); VIN1/MODE/PR1 peaks are bounded by U24. Unplug with a cell: +5V follows
 VBUS to the PR1 threshold, then the 8 u break-before-make gap: dV = 0.74 V x 8 u / (C12 + C13 + C15, 3.44 uF each at 3.4
 V, Murata curve) = 0.55 V → +5V = 3.19 V typical; worst corner (3.44 V, -20 % caps) 2.75 V. +3V3 stays >= 3.07 V: the
 P5B5 to current source is current and its 7 uF output cap carries 0.2 A for 8 u. V1 case (b) scopes it.
 C12 10 uF at VOUT: -5V hold-up. Mux 40/60 mOhm: can = 0.04 V at 0.70 A VIN1; HD about 0.25 A (the O4 is on +5V_BST ahead
 of the mux, D53 BR-10).
 Ti 7.3.4 (clamp VOUT or about 100 uF where RCB occurs): RCB happens by design at VIN1 vout/unplug with a cell; considered
 deviation, no part (short +5V capacitor, near-zero current at the 42-70 mV trip). V1: VIN1/VOUT <= 6.0 V.

SHEET NOTES

POWER sheet, generated with hardware_tools/sch_build.py + sch_root.py (P4 power sheet, do not edit by hand).
Ralls are power symbols (net = Value); +BATT_IN, +BATT, +5V_STBY, +5V_USB, +5V, +5V_CAM, +5V_HD, +3V3, GND. Sheet pins: HD_EN (in),
CURR Sense (out), VBUS (in). Local nets: SHUNT_Sense_P/N(CIN), U2_SW, U2_FB, U3_PR1, U5_EN.
Closed here: P4-1, P4-2, P4-3, P4-7, P4-9, P4-12 (+3V3), P4-17 (BR-14 C13), P4-19 (C30 deleted); R11 is the GPIO27 pull-down P4-4
counts on; P4-13 is void (D69: C2, C5, C14 stay XSR, sized on DC bias). Values: research/DESIGN-SPEC.md, 4.1, 15.5; P4 hand calculations
reproduced in the sheet notes.
Deleted: C17 (U5 CIN) (0.77 nF) + C30 (ESC3 EFM8 VDD bulk) deleted; each ESC now has one 10 uF hot-loop cap per AGM210MOP on its own sheet, and
ESC3's slot-M cap C35 sits within 4 mm of U8 VDD pin 4 (BR-05 as amended, 100 nF C31 at the pin). C17 (U5 CIN) deleted; C8-C11 serve, P5
placement condition above.

OpenDrone

Sheet: /POWER/
File: power.kicad.

Title: Power: input, current sense, 5 V, 3.3 V, 04

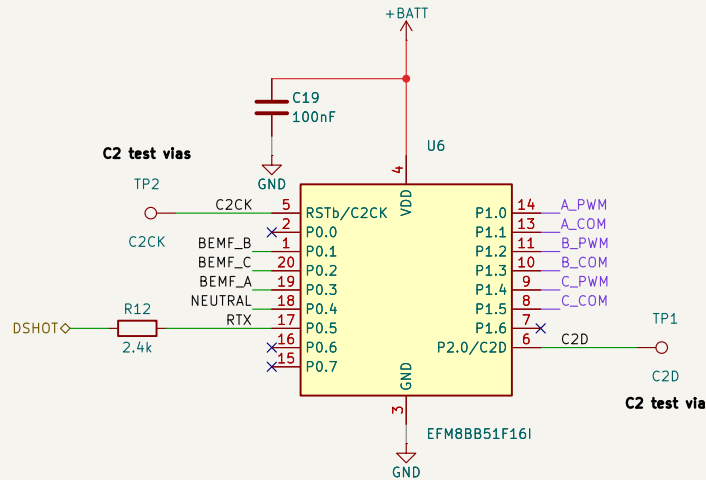
Size: A3

Date: 2026-10-09

Rev: rev1

d: 2/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A

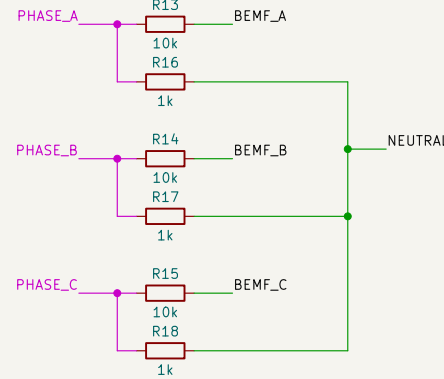


Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B_Mux, P0.2 C_Mux, P0.3 A_Mux, P0.4 V_Mux (neutral), P0.5 RTX; P1.0/P1.1 A_Pwm/A_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D. Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).

VDD pin 4 straight on +BATT, no series R (it must track the P sources); C19 100 nF X6S at pin 4 / GND pin 3. The data sheet's 1 uF (Fig. 5.1) is the phase B hot-loop cap C23 10 uF X6S (about 2.6–2.8 uF effective at 4.2 V, Murata DC-bias curve); BR-05 as amended (P4 critique): within 4 mm of pin 4 through the +BATT plane, 100 nF at the pin (v3 cell site 3.4–3.9 mm). ESC3 has no extra cap (C30 deleted, D77). VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7–5.25 V (V3c). Ramp: DS Table 4.7 RAMPVDD >= 10 us/V vs about 1 us/V at a hot plug (spec 4.1): Matrix-parity deviation, V3c 20 hot plugs; a 2.2–4.7 Ohm VDD R (tau 0.5 us) does not fix it.

DSHOT (P4–B): R12 2.4k at RTX. At a 2.5 V cell the 3.3 V DShot high sits up to 0.53 V over P0.5's VDD + 0.3 V abs max; R12 holds the clamp current to <= 0.35 mA ((3.33 – 2.5) V / 2.4k; about 0.1–0.2 mA after the clamp drop) against 50 mA per pin. RSTb/C2CK: internal pull-up only (PINMAP 12). C2D / C2CK: TP1 / TP2, filled test vias (D67), bench recovery only: NextPCB pre-programs Bluejay; updates via Betaflight passthrough.

BEMF network: 10k series + 1k neutral star



Source: tinyPEPPER single_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF_A P0.3, BEMF_B P0.1, BEMF_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to < 0.1 mA, 1k star, worst case one phase low and two high: 2/3 x 5.25 V / 1k = 3.5 mA, 12 mW per 0201 (50 mW part). P4–B: the netlist puts BEMF_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A_Mux 3 / B_Mux 1 / C_Mux 2 / V_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom_plan.json:

ESC1 M1 rear right: U6 C19, hot loop C20 C23 C21, R12–R18, Q29–Q31, TP1/TP2

ESC2 M2 front right: U7 C25, hot loop C26 C29 C27, R19–R25, Q32–Q34, TP3/TP4

ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26–R32, Q35–Q37, TP5/TP6

ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33–R39, Q38–Q40, TP7/TP8

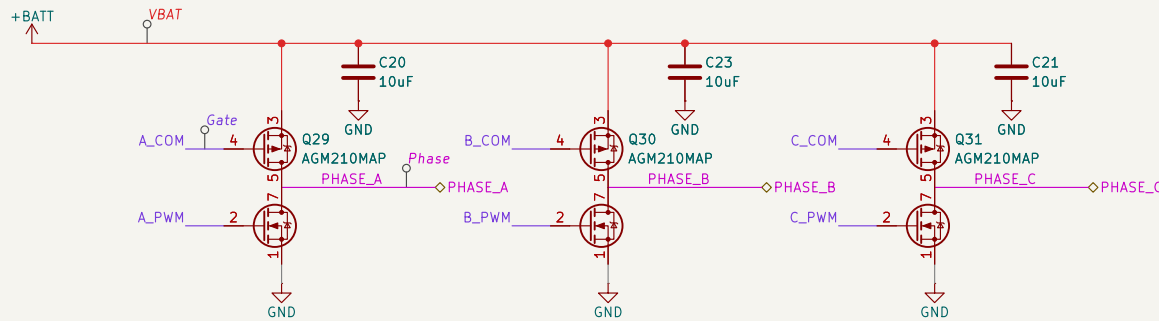
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE_x, motor lands J4/J7/J10/J13). In-sheet nets: x_COM, x_PWM (Gate class), BEMF_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE_A), Gate (A_COM), VBAT (+BATT rail): setup_board.py patterns /ESC?/PHASE_?, /ESC?/?_COM, /ESC?/?_PWM assign the rest.

Source: copied from OpenAIO hardware/esc_channel.kicad_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4–11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR <= 390 us) and the Bluejay bootloader turns on the 70–220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1–2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through >= 72k (pull-up + R12), i.e. <= 60 uA: 0.72 V over the 3.63 V unpowered–FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the PIO DShot idle–high back–feeds +BATT through the EFM8 I/O diode, <= (3.3 V – VF) / 2.4k, about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view: D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS +12 V, VDS 20 V abs max vs VDD <= 5.25 V regen peak. Hot loop (D77, P4 critique): one cap per AGM210MAP, C20 / C23 / C21 10 uF X6S 0402 from pin 3 S2 (+BATT) to pin 1 S1 (GND) at the cell's lead side, both pads via-in-pad, so the commutation loop closes at the pins (Matrix II: one cap per FET); the three are also the ESC +BATT bulk, Bluejay: first power-up A_X_10_96 (about 200 ns; DS td(off) + tf = 58 ns N at RGEN 3.3 Ohm, 84 ns P at 1 Ohm, longer on EFM8 port drive), then A_X_5_96 (Matrix II) after the V2 shoot-through check (PINMAP 7.3); the DS Qg of 23 / 33 nC disagrees 3–5x with Ciss 950 pF, so V2 measures the gate edges.

OpenDrone

Sheet: /ESC1/

File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

Size: A4

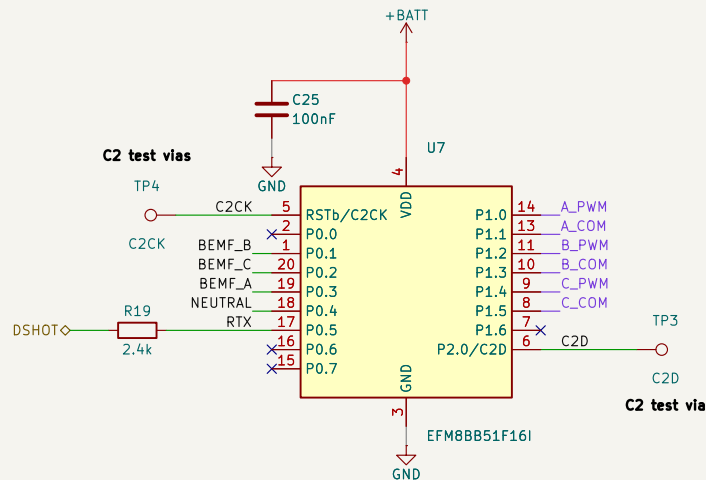
Date: 2026–10–09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 3/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A

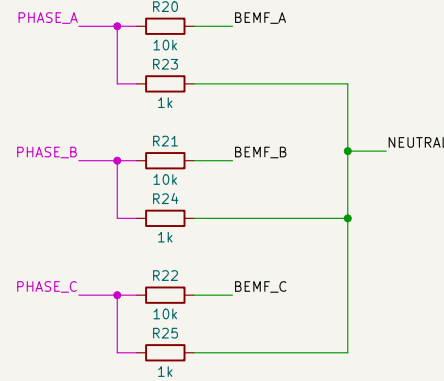


Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B_Mux, P0.2 C_Mux, P0.3 A_Mux, P0.4 V_Mux (neutral), P0.5 RTX; P1.0/P1.1 A_Pwm/A_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D. Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).

VDD pin 4 straight on +BATT, no series R (it must track the P sources): C19 100 nF X6S at pin 4 / VDD pin 3. The data sheet's 1 uF (Fig. 5.1) is the phase B hot-loop cap C23 10 uF X6S (about 2.6–2.8 uF effective at 4.2 V, Murata DC-bias curve); BR-05 as amended (P4 critique): within 4 mm of pin 4 through the +BATT plane, 100 nF at the pin (v3 cell site 3.4–3.9 mm). ESC3 has no extra cap (C30 deleted, D77). VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7–5.25 V (V3c). Ramp: DS Table 4.7 RAMPVDD >= 10 us/V vs about 1 us/V at a hot plug (spec 4.1): Matrix-parity deviation, V3c 20 hot plugs; a 2.2–4.7 Ohm VDD R (tau 0.5 us) does not fix it.

DSHOT (P4–B): R12 2.4k at RTX. At a 2.5 V cell the 3.3 V DShot high sits up to 0.53 V over P0.5's VDD + 0.3 V abs max; R12 holds the clamp current to <= 0.35 mA ((3.33 – 2.5) V / 2.4k; about 0.1–0.2 mA after the clamp drop) against 50 mA per pin. RSTB/C2CK: internal pull-up only (PINMAP 12). C2D / C2CK: TP1 / TP2, filled test vias (D67), bench recovery only: NextPCB pre-programs Bluejay; updates via Betaflight passthrough.

BEMF network: 10k series + 1k neutral star



Source: tinyPEPPER single_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF_A P0.3, BEMF_B P0.1, BEMF_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to < 0.1 mA, 1k star, worst case one phase low and two high: 2/3 x 5.25 V / 1k = 3.5 mA, 12 mW per 0201 (50 mW part). P4–B: the netlist puts BEMF_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A_Mux 3 / B_Mux 1 / C_Mux 2 / V_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom_plan.json:

ESC1 M1 rear right: U6 C19, hot loop C20 C23 C21, R12–R18, Q29–Q31, TP1/TP2

ESC2 M2 front right: U7 C25, hot loop C26 C29 C27, R19–R25, Q32–Q34, TP3/TP4

ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26–R32, Q35–Q37, TP5/TP6

ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33–R39, Q38–Q40, TP7/TP8

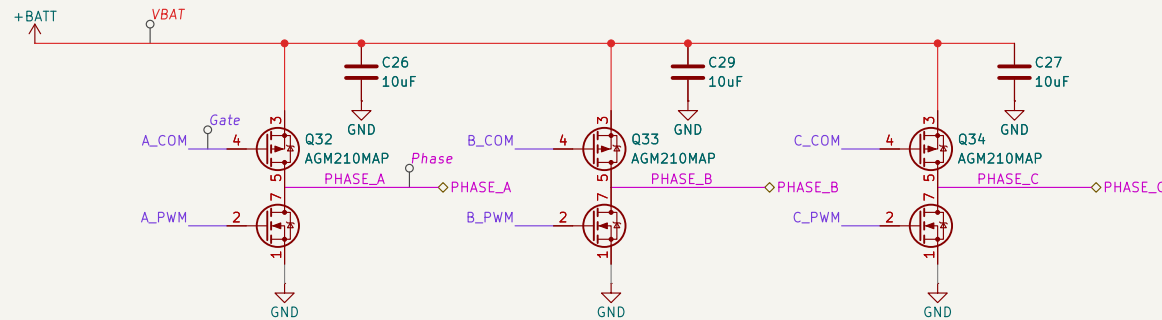
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE_x, motor lands J4/J7/J10/J13). In-sheet nets: x_COM, x_PWM (Gate class), BEMF_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE_A), Gate (A_COM), VBAT (+BATT rail): setup_board.py patterns /ESC?/PHASE_?, /ESC?/?_COM, /ESC?/?_PWM assign the rest.

Source: copied from OpenAIO hardware/esc_channel.kicad_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4–11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR <= 390 us) and the Bluejay bootloader turns on the 70–220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1–2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through >= 72k (pull-up + R12), i.e. <= 60 uA: 0.72 V over the 3.63 V unpowered–FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered): V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the PIO DShot idle–high back–feeds +BATT through the EFM8 I/O diode, <= (3.3 V – VF) / 2.4k, about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view: D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS +12 V, VDS 20 V abs max vs VDD <= 5.25 V regen peak. Hot loop (D77, P4 critique): one cap per AGM210MAP, C20 / C23 / C21 10 uF X6S 0402 from pin 3 S2 (+BATT) to pin 1 S1 (GND) at the cell's lead side, both pads via-in-pad, so the commutation loop closes at the pins (Matrix II: one cap per FET); the three are also the ESC +BATT bulk, Bluejay: first power-up A_X_10_96 (about 200 ns; DS td(off) + tf = 58 ns N at RGEN 3.3 Ohm, 84 ns P at 1 Ohm, longer on EFM8 port drive), then A_X_5_96 (Matrix II) after the V2 shoot-through check (PINMAP 7.3); the DS Qg of 23 / 33 nC disagrees 3–5x with Ciss 950 pF, so V2 measures the gate edges.

OpenDrone

Sheet: /ESC2/

File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

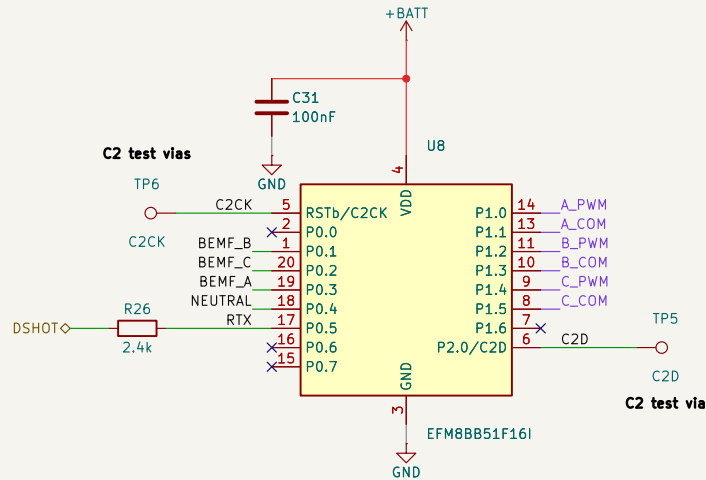
Size: A4 Date: 2026–10–09

KiCad E.D.A. 10.0.6

Rev: rev1

Id: 4/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A

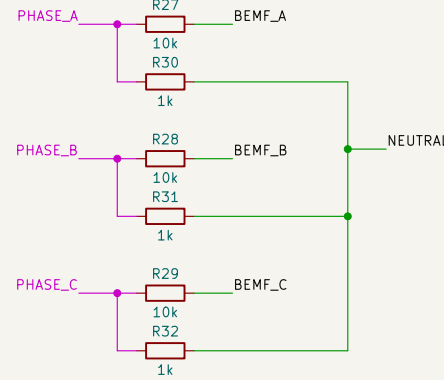


Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B_Mux, P0.2 C_Mux, P0.3 A_Mux, P0.4 V_Mux (neutral), P0.5 RTX, P1.0/P1.1 A_Pwm/A_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D. Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).

VDD pin 4 straight on +BATT, no series R (it must track the P sources): C19 100 nF X6S at pin 4 / GND pin 3. The data sheet's 1 uF (Fig. 5.1) is the phase B hot-loop cap C23 10 uF X6S (about 2.6–2.8 uF effective at 4.2 V, Murata DC-bias curve); BR-05 as amended (P4 critique): within 4 mm of pin 4 through the +BATT plane, 100 nF at the pin (v3 cell site 3.4–3.9 mm). ESC3 has no extra cap (C30 deleted, D77). VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7–5.25 V (V3c). Ramp: DS Table 4.7 RAMPVDD >= 10 us/V vs about 1 us/V at a hot plug (spec 4.1): Matrix-parity deviation, V3c 20 hot plugs; a 2.2–4.7 Ohm VDD R (tau 0.5 us) does not fix it.

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ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26–R32, Q35–Q37, TP5/TP6

ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33–R39, Q38–Q40, TP7/TP8

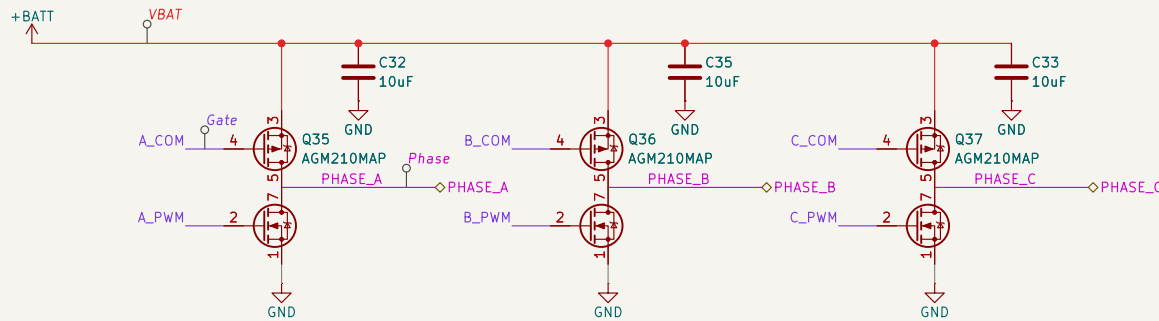
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE_x, motor lands J4/J7/J10/J13). In-sheet nets: x_COM, x_PWM (Gate class), BEMF_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE_A), Gate (A_COM), VBAT (+BATT rail): setup_board.py patterns /ESC?/?/PHASE_?, /ESC?/?_COM, /ESC?/?_PWM assign the rest.

Source: copied from OpenAIO hardware/esc_channel.kicad_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4–11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR <= 390 us) and the Bluejay bootloader turns on the 70–220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1–2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through >= 72k (pull-up + R12), i.e. <= 60 uA: 0.72 V over the 3.63 V unpowered–FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered): V9 scopes the pad at power-up; fallback a 3.3 V–domain clamp or a DShot-path enable (not in rev1).

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OpenDrone

Sheet: /ESC3/

File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

Size: A4

Date: 2026–10–09

Rev: rev1

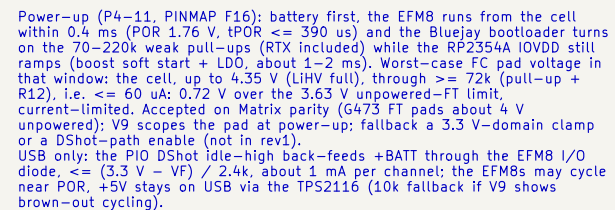
KiCad E.D.A. 10.0.6

Id: 5/14

5



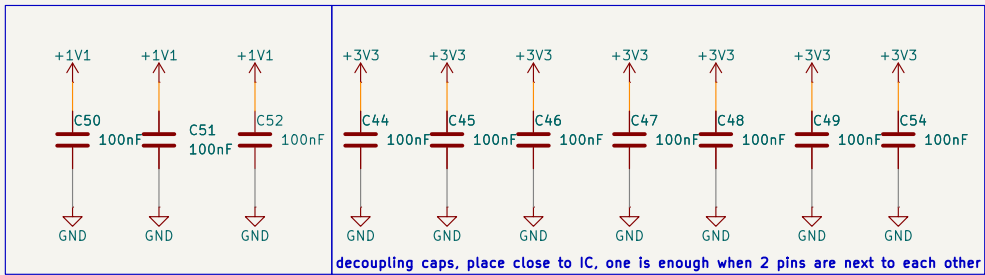
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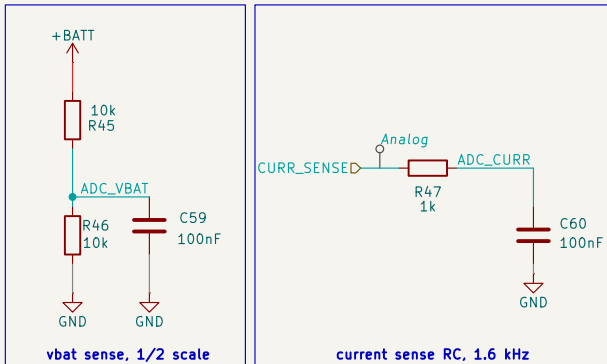
OpenDrone

Id: 6/14

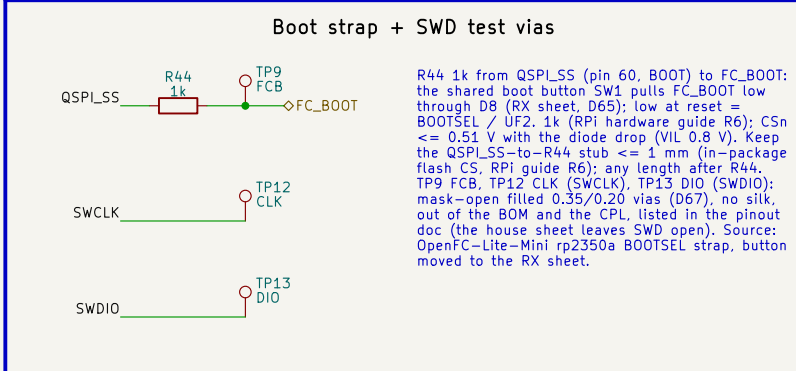
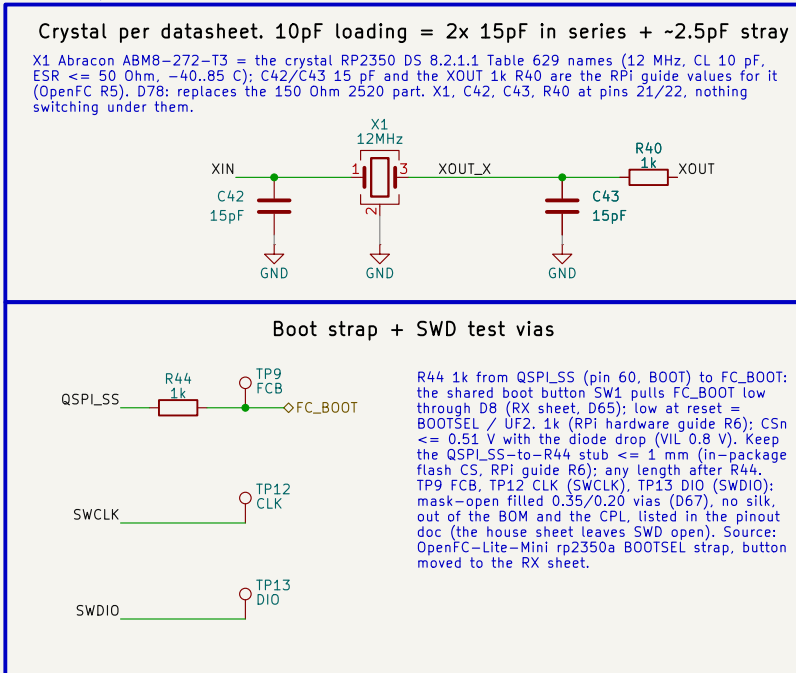
RP2354A: Smaller variant with integrated flash (2MB)



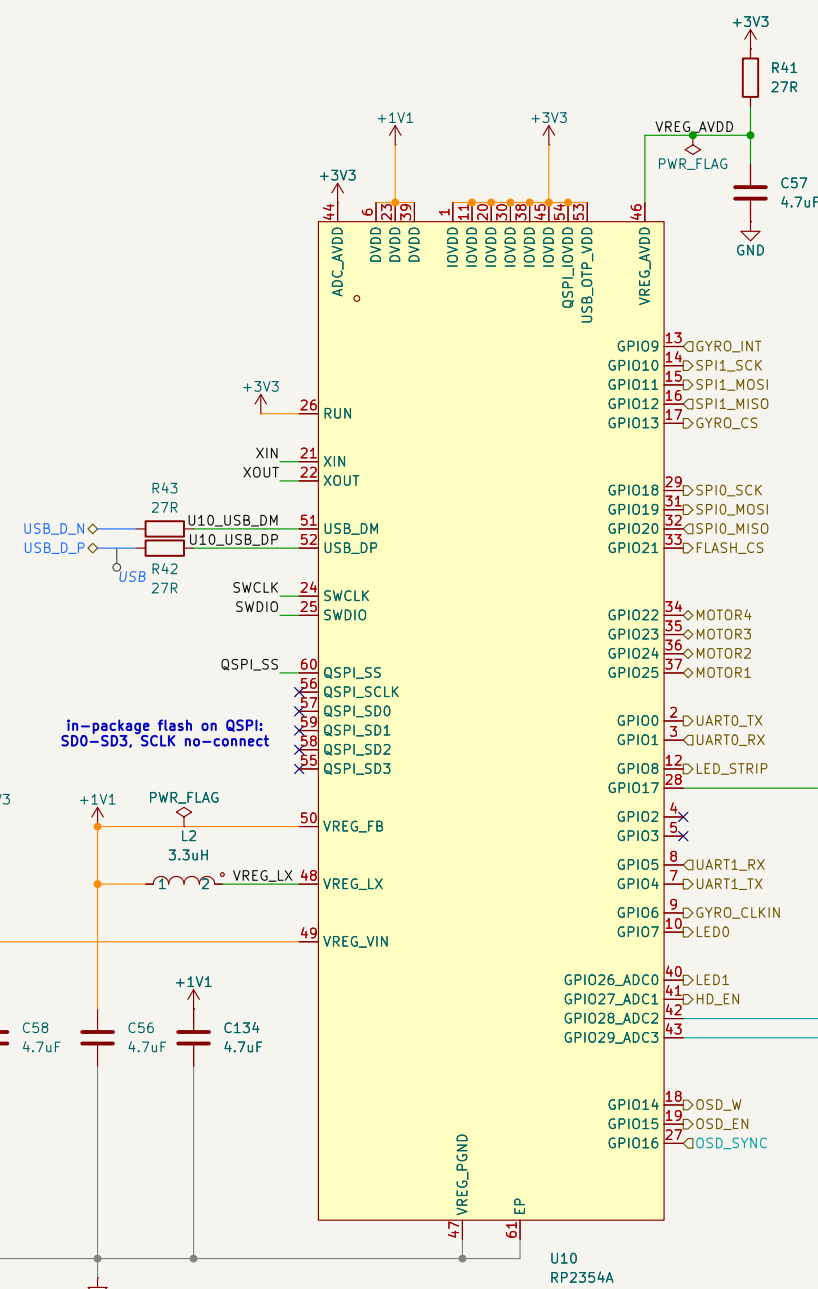
DVDD 6/23/39: C50–C52. IOVDD 1/11/20/30/38: C44–C48, one per pin. C49 sits at ADC_AVDD pin 44 and also serves IOVDD 45 (BR–08: RP2350 DS 6.1.5: 100 nF close to ADC_AVDD). C54 <= 1 mm from QSPL_IOVDD pin 54 (the in-package flash supply, 2.97–3.63 V) and also serves USB_OTP_VDD 53 (BR–08, P4–17 placement condition). All Murata GRM033C61E104KE14D 100 nF X6S 0201 (BR–18). Copied from OpenFC–Lite–Mini rp2350a (7 caps for the 9 3V3 pins there too); its 4.7 uF on this row moved to the SMP5 block (C56).



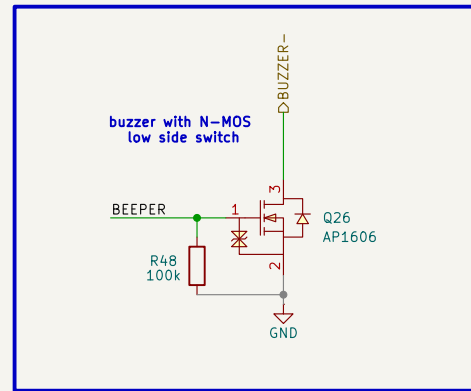
VBAT: R45/R46 10k/10k (house 100k/10k), 4.35 V => 2.18 V on GPIO29 (ADC3, not 5 V tolerant). vbat_scale 200/100; 5 kOhm with C59 = 318 Hz. CURR: R47 1k + C60 100 nF = 1.6 kHz on GPIO28 (ADC2); INA186A3 on POWER; 50 mV/A. Ibatascale 500. Copied from OpenFC–Lite–Mini rp2350a (vbat + ESC current).



R42/R43 27R within 1–2 mm of pins 52/51 (RPI HW design 5.1, house 30R). USB_D_P/N: 90 ohm pair over unbroken GND, length-matched (D53); D7 ESD + J31 on PADS. Pin-side nets: U10_USB_DP/DM.



RP2350 1.1V DC–DC SMP5: follow the datasheet placement; RP2350 DS 6.3.8 (binding): C58 = CIN at VREG_VIN, C56 = COUT on +1V1 with VREG_FB taken from the C56 pad (not under L2); C58, L2 and C56 on the U10 side, smallest hot loops; CIN/COUT GND to VREG_PGND at one point with 2 vias; no copper under L2 / VREG_LX on the next layer. L2 AD1A–B201610S3R3 is polarity-marked; orientation as DS Fig. 26 / 28 (pin 2 = VREG_LX as on the flown house board). P5 condition (tools/check_p5_conditions.py SMP5): C58, L2, C56 and C57 on U10's side within 3 mm of pins 49 / 48 / 50 / 46. R41 27R + C57 4.7 uF filter VREG_AVDD (1.25 kHz, BB–21: 3.135–3.63 V); C57 gets its own GND via to the EP. D5: CIN >= 4.7 uF, COUT 4.7 uF +/-20 %; C134 = the 2nd 4.7 uF on +1V1 that DS 6.3.8 recommends at DVDD pin 23 (fitted per the 2026–10–09 ruling, house board fits two); place it at pin 23.



Q26 AP1606 (copied from OpenFC–Lite–Mini rp2350a, Q5 there); 20 V, 0.7 A, Vgs(th) 0.4–1.2 V, RDS(on) <= 220 mOhm at Vgs 2.5 V (ALLPOWER DS p2). GPIO17 high = on (BEEPER_INVERTED, active buzzer, no PWM); R48 100k holds the gate low through reset. BZ+ = +5V and BZ– = drain on the GND LED 5V BZ– pad row (PADS).

gyro SPI1 (IMU: GYRO_CS pull–up R49 there)

blackbox SPI0 (FLASH_CS pull–up R50 there)

PI00 bidir, DShot to the EFM8s (2.4k at the EFM8)

for CRSF receiver (ESP32 ELRS, UART0)

WS2812 pad LED, driven direct (F1, BR–04)

GPIO2/3 spare, no–connect (D89)

for MSP DisplayPort: DJI O4 Lite TX1/RX1

gyro CLKIN (ICM–42688–P only)

status LEDs (LED sheet, sinking)

HD line PINIO1: high = 04 on, analog VTX off

PIO2 framebuffer OSD: W, EN, SYNC consecutive

SHEET RP2350A, source: OpenFC–Lite–Mini/hardware/rp2350a.kicad_sch (flown Rev2), copied block by block with its layout, frames and text style. Changes: USB–C + CC removed (USB = JST SH J31 + DT on PADS; 27R series R42/R43 stay at U10, D52 BR–21 value); I2C0 pull–ups removed (GPIO4/5 = UART1); the inverting WS2812 stage is not copied (F1, BR–04; GPIO8 drives the LED pad); status LEDs moved to the LED sheet; BOOTSEL button replaced by R44 + FC_BOOT (shared button on RX); SWD to test vias (D67); VBAT divider 10k/10k; 12 Mhz crystal = ABM8–272–13 (D78); VREG_AVDD 27R; two 4.7 uF on +1V1 (C56 at VREG_FB, C134 at DVDD 23). GPIO map: research/PINMAP.md 2–3 (30 of 30 GPIOs, no move); Betaflight config PINMAP 5.

P4–11 power–up and 5 V–tolerant pads (PINMAP 4.5, F16; RP2350 DS Table 1433); an FT pad takes <= 3.63 V while IOVDD is 0–1.8 V (4.2 V at 2.5 V, 5.5 V at 3.3 V); the ADC pads GPIO26–29 take <= IOVDD + 0.5 V. – MOTOR1–4 (GPIO25–22, F1), battery first: each EFM8 weak pull–up (70–220 kOhm to VDD = cell) reaches the pad through its 2.4k before +3V3 is up (boost soft start + LP5912 ramp, about 1–2 ms). Worst case 4.35 V open circuit = 0.72 V over 3.63 V, from >= 72 kOhm (<= 60 uA). Matrix parity (G473 FT pins –4 V at VDD 0). V9 scopes it: fallback (F16): 3.3 V–domain clamp or DShot–path enable. – ADC_VBAT (GPIO29): R45/R46 put up to 2.18 V on the pad from 5 kOhm while IOVDD < 1.7 V, so the pad clamp to IOVDD conducts <= 0.3 mA during the ramp; House + Matrix parity; V9 scope. – HD_EN (GPIO27): R11 2.4k + the BA516LD from O4_EN (<= 90 uA) hold it at <= 0.22 V before and during reset; inside IOVDD + 0.5 V and below the AP1606 HD–gate Vgs(th) min 0.4 V at 25 C (P4–4: about 0.25 V at 85 C, –2.5 mV/K; a hot low–Vth part may hold the analog VTX off until PINIO1 is driven low, R68 47k, V9 hot–soak boot check); driven high it is >= 2.62 V (VOH, 1.4 mA into R11) over Vgs(th) max 1.2 V. – UART0, SPI0/1, gyro, OSD, CURR_SENSE: every driver is on +3V3 = IOVDD, no excursion. UART1 TX1/RX1: the O4 is fed only through U5 (needs HD_EN high), none. USB D+/D–: the host drives only after the device pull–up. – TX1/RX1 used as the free hardware UART in analog builds (IO–SET row B; the only free UART since D89); a peripheral on a 5V pad is up before IOVDD: 3.3 V logic only (README); a 5 V TX on RX1 (GPIO5) would exceed 3.63 V at power–up. – GPIO2/3 (pins 4/5): no–connect since D89 (no user UART pads).

OpenDrone

Sheet: /RP2350A/
File: rp2350a.kicad_sch

Title: RP2354A flight controller MCU

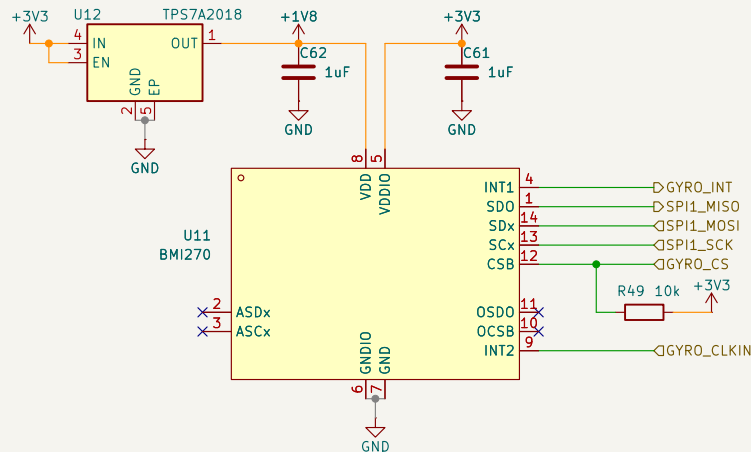
Size: A2 Date: 2026–10–09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 7/14

Universal IMU wiring and LGA-14 land (BMI270, ICM-42688-P)



GYRO BMI270: U11, R49

Source: OpenFC-Lite-Mini imu.kicad_sch (universal LGA-14 land, SPI1 4-wire, R49), copied and adapted. BMI270 on the BOM; TDK ICM-42688-P qualified on the same land (V10 fly-off, one part per revision).

Pins 2/3 (BMI270 ASDx/ASCx, TDK RESV) and 10/11 (OCSB/OSD0, TDK RESV): no-connect. BMI270 DS Table 22: unused ASDx/ASCx go to VDDIO or stay open, never to GND (the house sheet grounded them for the LSM6DSV16X).

Pins 6/7 GND (TDK pin 7 RESV must be GND). VDDIO pin 5 = +3V3 (RP2354A I/O level). VDD pin 8 = +1V8; the BMI270 has no supply-order rule (DS 4.3). Abs max 4.0 V on VDD/VDDIO, I/O up to VDDIO + 0.3 V.

Pin 9: TDK CLKIN = GYRO_CLKIN (RP2354A GPIO6, PWM 32 kHz, driven only with the TDK part); BMI270 INT2, unused, left disabled. R49 10k: GPIO13 resets with its pull-down on, which would select the chip; R49 holds CSB >= 2.58 V (0.78 x VDDIO with the RP2354A RPD min 36 kOhm at IOVDD 3.3 V, RP2350 DS Table 1683) > VIH 0.7 x VDDIO = 2.31 V.

P5: top side next to U12, LGA rotated 90 deg: GYRO_1_ALIGN from the placed orientation (PINMAP F11).

+1V8 GYRO LDO: U12, C61, C62 (D52 BR-09, P4-17)

Source: pcb-agent-commons blocks/LDO_IMU_TPS7A2033 (OpenFC-H7) topology with the 1.8 V part and no ferrite (fed from the +3V3 LDO): the house OpenFC-Lite-Mini also runs the gyro VDD from its own 1.8 V LDO. TPS7A2018 DQN: 1 OUT, 2 GND, 3 EN, 4 IN, 5 EP to GND. IN = EN = +3V3: VIN 1.6-6.0 V, VEN 0-6.0 V (abs max 6.5 V), EN never above IN. Load < 1 mA.

No 100 nF at the gyro: C61 = U12 CIN at BMI270 VDDIO pin 5, C62 = U12 COUT at VDD pin 8 (COUT >= 0.47 uF effective, ESR <= 100 mOhm: about 0.6-0.85 uF at 1.8 V). C61 counts in the +3V3 COUT sum (P4-12, POWER).

P5: U12 on the gyro side with OUT facing pin 8; C61 between U12 pin 4 and gyro pin 5, C62 <= 1.5 mm from pin 8 (P5-06), else re-place U12.

OpenDrone

Sheet: /IMU/

File: imu.kicad_sch

Title: Gyro BMI270 + 1.8 V LDO

Size: A5

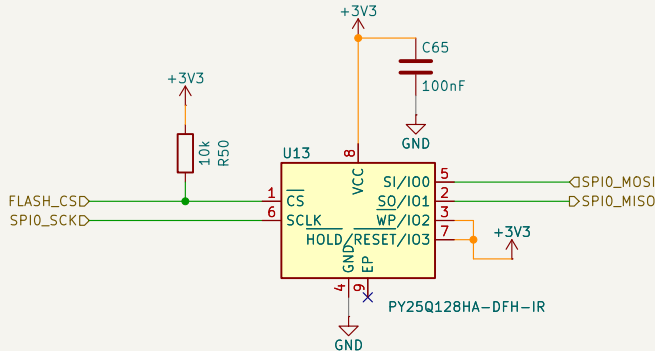
Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 8/14

SPI NOR PY25Q128HA: 16 MB blackbox on SPI0 (U13 C65 R50)



Placement (P5): beside the RP2354A on SPI0 (bom_plan region nor, top; either side works). U13, SPI0 and FLASH_CS stay ≥ 3 mm from the RX antenna hole (spec 4.9 and 11, DRU 'SPI0 3 mm from the RX antenna hole'): SPI0 harmonics fall in the ELRS band, V7 measures desense while logging. C65 at pin 8. Filled+capped 0.35/0.20 via-in-pad allowed in the lead lands, never in the tabs. Betaflight sets SCK to fast slew (bus_spi_pico.c); MOSI and CS keep the reset 4 mA drive and slow slew (EMI lever without parts). Sourcing: no LCSC listing; NextPCB buys by MPN (D60), the quote confirms it. Same-pinout fallbacks in the Betaflight table (USON-8 4x4 land not staged): PY25Q128HA-QVH-IR, GD25Q128EQIGR.

Source: OpenFC-Lite-Mini hardware/blackbox.kicad_sch (rev2). Kept: layout, the 10k CS pull-up (R39 $\rightarrow$ R50) and the 100n supply cap (C43 $\rightarrow$ C65). Changed: its microSD (Card1) became this 16 MB SPI NOR (D66); the SPI0(SCK,MOSI,MISO) bus label became four hierarchical labels (sch_contract.json).

U13 Puya PY25Q128HA-DFH-IR: 128 Mbit, USON-8 3x3 (DF), 2.7-3.6 V, -40..85 C, fC 133 MHz, READ 03h 80 MHz, JEDEC ID 85 20 18 is in the Betaflight m25p16 table: USE_FLASH_M25P16, SPI0 PA18/PA19/PA20, FLASH_CS_PIN PA21 (PINMAP). SPI0 runs at 75 MHz for writes AND reads: Betaflight bus_spi_pico.c spiCalculateDivider() uses an even prescale ≥ 2 on clk_peri 150 MHz, so the table's 133 / 80 MHz both give 150/2. Writes are inside fC; reads are not guaranteed: tCLQV 6-7 ns vs the 6.67 ns SCK half period plus pad and trace delay (the RP2350 DS gives no SPI receive timing). Gate F7 / V9: erase, write, read back and CRC a full log, status polling under load; fallback = a lower read clock for this JEDEC ID (firmware cap or table entry), upstreamed with the board config.

WP# (3) and HOLD#/RESET# (7) tied to +3V3: the DS gives an internal pull-up only to the SOP-16 dedicated RESET#, so pin 7 must not float. Delivered SR = 00h, CR = 00h (QE 0, HOLD# function, WPS 0: no block lock) and Betaflight sends 1-bit SPI commands only, so I02/I03 never drive against the rail (the ERC pin_to_pin warning on pins 3/7 against a +3V3 power output is this deliberate tie).

R50 10k: CS# must follow VCC at power-up (DS 5.5). RP2354A pads reset with the pull-down on (PDE 1, RPD 36-113 kOhm at 3.3 V): CS $\geq 36/46$ VCC = 0.78 VCC > VIH 0.7 VCC, so U13 stays deselected through reset and boot. U13 and the RP2354A IOVDD share +3V3: no pin above VCC + 0.5 V.

C65 100 nF X65 at pin 8, GND return straight to pin 4 (ICC max 13 mA read, 12 mA erase; adds 0.1 uF nominal to the +3V3 COUT sum of P4-12). EP (9) = two tie-bar tabs with no electrical function in the DS: no-connect, no vias in the tabs.

OpenDrone

Sheet: /BLACKBOX/

File: blackbox.kicad_sch

Title: Blackbox 16 MB SPI NOR PY25Q128HA

Size: A5

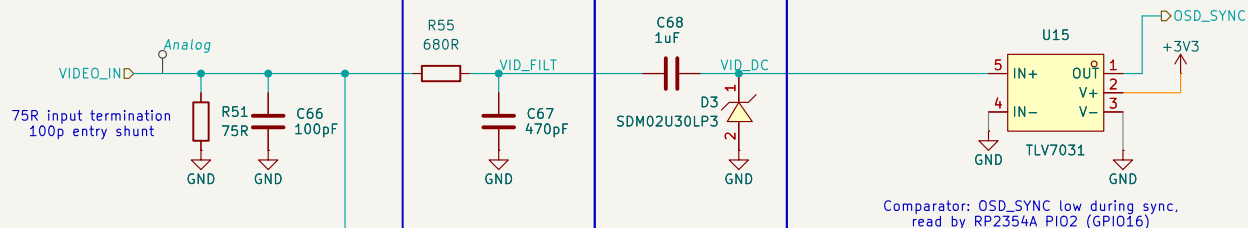
Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 9/14

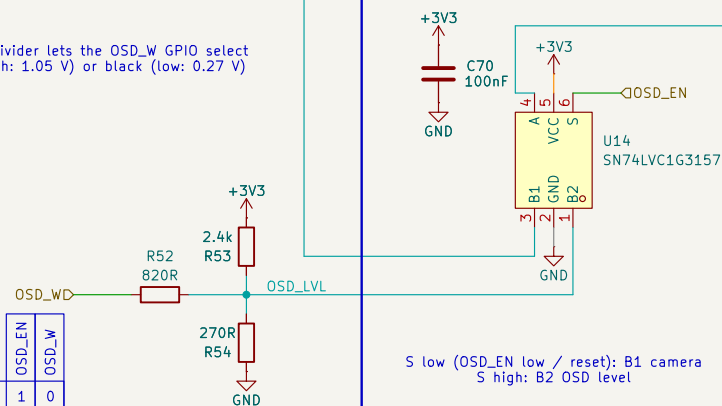
VIDEO PATH



U15 V+ (2) is decoupled by C70 in the switch cell,
one 100n shared with U14 (BR-13)

Comparator: OSD_SYNC low during sync,
read by RP2354A PI02 (GPIO16)

Resistor divider lets the OSD_W GPIO select white (high: 1.05 V) or black (low: 0.27 V)



S low (OSD_EN low / reset): B1 camera
S high: B2 OSD level

Analog switch

VIDEO_OUT: the switch output drives the RTC6705 video network on the VTX sheet directly (R58 1.2k $\parallel$ C104 33p, R59 270R, C105 4.7u, R60 1.2k, OpenOSD_X values); the house COS8051 buffer (U1, R46-R48, C41) is deleted (spec 4.8).

OSD levels (P4-10, recomputed at the P4 critique with the real load): open-circuit at OSD_LVL (+3V3 3.30 V, OSD_W push-pull) black 0.268 V / white 1.054 V from 195R (R53) | R54 = 0.352 V / 256R, with R52 + switch on 7-18R. VIDEO_OUT is loaded by the VTX network ahead of C105: R58 + R59 = 1.47 kOhm at DC, about 0.95 kOhm at 5 MHz and 0.60 kOhm at 10 MHz (R58 / C104 4.0 MHz, 21.9 MHz with R59; 270R only about 20 MHz). Loaded levels: black 0.234-0.236 V, white 0.920-0.927 V (swing x0.88 of the open-circuit 0.786 V); edge gain x0.86 at 5 MHz and x0.80 at 10 MHz vs x0.97 for the camera (37.5R source); OSD text slightly dimmer and softer than the camera picture, not a failure (the house COS8051 buffer then checks the source low-impedance is deleted). VS checks OSD white + black and edge sharpness on a monitor; fallback: rescale R52-R54 within the 3 mA OSD_W budget. PS-01 (R54 270R) would lower white to about 0.89 V: not recommended.

OSD_W: 2.74 mA high ($\leq$ 3 mA, 2.79 mA at 3.36 V), 0.33 mA low.

Source: OpenFC-Lite-Mini hardware/osd.kicad_sch (rev2, flown), copied and adapted: cells, texts, truth table and part spots kept (A3 → A4). Same topology: 75R termination, 680R/470pF chroma filter, 1u + SMD02U30LP3 sync-tip clamp, TLV7031 comparator, SN74LVC1G3157 camera/level switch (contract source, spec 4.8).

```
Changed: COS8051 bffer deleted (output = VIDEO_OUT); divider R42 3.9k / R43
12k / R44 1.4k -> R52 820R / R53 2.4k / R54 287R (about 5x down, P4-10);
C39 + C42 -> one shared C70 100n X65 (D52 BR-13 + BR-18); C66 100p entry
shunt added (spec 11); refs per bom_plan (R41>R51, C38>C67, C40>C68, D9>D3,
U2>U15, R45>R55, R42-44>R52-54, U18>U14).
```

Symbols from the OpenDrone catalogue with TI pin numbers: TLV7031 X2SON 1 OUT, 2 V+, 3 V- (centre pad), 4 IN-, 5 IN+; SN74LVC1G3157 DTB 1 B2, 2 GND, 3 B1, 4 A, 5 VCC, 6 S. The house lib symbols used the EasyEDA pad order (consistent with its own footprints), so the drawing is re-routed with the same nets: IN+ = VID_DC, IN- = GND, B1 = camera, B2 = OSD level. The house sheet's rotated ground is upright here.

Operation: OSD_EN low (RP2354A reset pull-down, OSD idle) -> B1, the camera passes through. OSD_EN high -> B2: OSD_W high = white, low = black. No OSD without camera sync (osd_pico.c overlays on detected sync). OSD_W/EN/SYNC = GPIO14/15/16, consecutive, P102 (PINMAP).

Sync threshold (P4.0): 680R/470P = 498 kHz: colour burst 0.11–0.14 (–17–21 mV), 4.7 uS sync kept (tau 0.32 uS). C68 + D3 (cathode on VID_DC) clamp the sync tips to –VF; the leakage-limited clamp current (IA average, tens of uA at the tip) gives VF about 0.1–0.2 V over temperature (!): DS 0.37 V max at 10 mA / 25 °C, 0.20 V at 10 mA / 125 °C). IN+ at GND puts the trip point VF above the tip, 33–67 % of the 0.3 V sync, >= 0.08 V from tip and blanking after VIO +8 mV and VHY5 2–17 mV. OSD_SYNC is low during sync (osd_sync.wait for a low sync pin). IN+ at –VF stays inside abs max (VEE – 0.3 V), at most 0.1 V under the ROC input floor (VEE – 0.1 V), no phase reversal (DS 6.4.1). tpd 3 uS on both edges (equal shift, as on the house board).

Ratings: +3V3 only (U14 1.65-5.5 V, U15 1.6-6.5 V), S: VIH 0.7 VCC = 2.31 V, edges <= 10 ns/V (fast slew pulses). AC-coupled camera sync tips (~0.3 to ~0.4 V) are below the switch ROC (0..VCC) but inside abs max (~0.5 V, 50 mA; 37.5R source), as on the house board; V9 checks. Supply: U15 0.9 uA, U14 <= 10 uA, divider 0.94-1.26 mA + OSD-W 2.74 mA. C68 X5R carries <= 0.6 V DC: no DC-bias loss (P4-13 hot-zone rule void, D69).

Placement (P5: P4-17 sheet note): bottom, region osd, BR-13: C70 is the only supply cap of U14 (VCC 5) and U15 (V+ 2): place it between the two parts (2,5 mm apart), GND via-in-pad: the P5 check lists its distance to both pins. U15 centre pad (3, V-) to GND through a via (TI 7.4). R51 and C66 at the video entry on the CAM pad J23 / plug J32 line: camera returns into L2 at the CAM GND pad: video on L1/L6 over a plane with GND both sides, never L3 (spec 11). R55, C67, C68, D3 next to U15 IN+. Video parts away from the boost and its inductor (preference, D69: v2 floorplan 3.3-8.6 mm, video test).

Net3-class Analog: VIDEO (flag here), VIDEO_OUT, VID_FILT, VID_CD, OSD_LVL, OSD_SYNC by the setup_board.py patterns: OSD_W, OSD_EN Default. source: from N. MPN from (bom plan): LCSS empty where bom plan has one (C66, R52, R54) open item A9-LCSS, P5-01 (R54) RC0201R-07270RL shared with R59, -1 line): black 0.258 V, white 1.011 V, 187R, OSD_W 2.79 mA, also passes P4-10: proposed in contract-issues, drawn = bom-plan.

Sheet: /OSD/
File: osd.kicad_sch

Title: PIO analog OSD front end

Size: A4	Date: 2026-10-09
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KiCad E.D.A. 10.0.6

Rev: rev1

Id: 10/14

OpenDrone

[illegible]

ELRS RGB LED

D4 XINGLIGHT XL-1010RGBC-2812B (copied from OpenRX-Lite-UFL) on +3V3, ELRS led_rgb GPIO22, one LED (overlay ledidx_rgb_vtx [0]).

P4-15 DECIDED: KEEP ON +3V3. Datasheet: VDD 3.5-5.5 V, DIN VIH $\geq$ 2.8 V, 5 mA per colour with a 0.6 V sink knee. On +3V3 the supply is 0.2 V under the table minimum (blue / green may run below 5 mA; Vf + knee is about 3.4-3.5 V); DIN comes from a 3.3 V CMOS output on the same rail. On +5V the supply would be in range but the data guarantee is short (ESP32 VOH $\geq$ 0.8 VDD = 2.64 V at a high-impedance load, DS Table 18, vs VIH 2.8 V). Neither option sits fully inside the tables; the house RX circuit (OpenRX-Lite, -UFL, OpenAIO) runs this LED on 3.3 V, so it stays. Bench V-LED on the first boards: red, green, blue and white at 3.3 V, cold and hot. Fallback: VDD pin to +5V (one trace; C83 is 25 V).

Wi-Fi CHIP ANTENNA (D58)

Wi-Fi (D58): LNA-IN (2) -> RF_WIFI 50 Ohm line (<= 8 mm preferred, D69: FLOORPLAN open item 10) -> C132 shunt (DNP) / R88 0R series (fitted) / C133 shunt (DNP) -> AE2 Johanson 2450AT07A0100001T (FEED 1/4, GND 2/3). Same pi as Expressif Fig. 11 (C4/L1/C3) and OpenRX-Lite-UFL (C22/L2/C21). The PICO matches LNA_IN to 50 Ohm inside, so the network only tunes the antenna: values from V7 tuning (VNA at the feed or Johanson's service): R88 may become a Murata LQ6031N 0201 inductor. The AE2 footprint carries the 5.0 x 3.2 mm keep-out (DRU_D21 wifi_antenna_edge_strip).

NOTES

RX SHEET (P4 shard C). Sources: the SX1281 radio, 52 MHz TCXO, RGB LED and Wi-Fi pi match are COPIED from the house sheet `OpenRX-Lite-UFL hardware/esp32c3_sx1281_lite.kicad_sch` (same as `OpenAIO_rx_esp32c3_sx1281.kicad_sch`); same symbol positions and wires, adapted to `bom_plan refs/values/MPNs` and to `D52/D53/D58` (LDO mode, pin 5 to GND, UFL replaced by the wire antenna hole, `2450A0707A00001T` chip antenna, grounds turned down); the house `Johanson 2450F0M0700034T` front-end filter stays. The `ESP32-PICO` block is new, from the Espressif `ESP32-PICO` series datasheet v1.3 Fig. 11 (D52 BR-01, D55); the shared boot / bind button is new (D65, Matrix II style). Pin plan: `research/PINMAP.md` 6. Parts: `hardware/bom_plan.json`. Radio and LED +3V3 come from the `POWER` sheet (U4 LP5912-3.3).

BOOT / BIND BUTTON (D65)

BOOT / BIND BUTTON (D65; new). SW1 Alps SKUBAAE010 (2.4 x 1.4 mm) from BOOT_SW to GND. D8 Diodes SDM02M30CLP3 dual Schottky, common cathode on BOOT_SW; A1 = FC_BOOT (RP2354A QSPLSS through R44 1k, RP2350A sheet), A2 = RX_BOOT (ESP32 GPIO0, pin 23).

Released: each strap sees only diode leakage (IR <= 0.15 uA at 10 V, 25 C), so GPIO0 keeps its internal pull-up and QSPLSS its own level.

Held while power is applied: RP2354A BOOTSEL and ESP32 ROM download mode (CSn <= 0.51 V vs VIL 0.8 V; GPIO0 <= 0.42 V vs VIL 0.825 V); each mode stays latched until that chip's next reset.

Production / ELSR recovery (D67, spec 12.1): (1) hold SW1 (or probe BOOT_SW) while powering; (2) write the Betaflight UF2 (or picotool reboot); the RP2354A restarts into Betaflight, the ESP32 stays in download mode; (3) espotool through Betaflight serial passthrough.

Pressed while running: ELSR button (bind 1.5 s, Wi-Fi 5 s, reboot 12 s); 2.9 mA through R44, CS stays >= 2.62 V, the FC keeps running.

Bench V-btn: if GPIO0 dips below 2.5 V while released (QSPLSS edges under XIP load coupling through D8), add 100 pF 0201 on BOOT_SW.

ESP32-PICO-V3 MCU (ELRS)

ESP32-PICO-V3 (U16), Espressif PICO datasheet v1.3 Fig. 11 / Table 4 (new block).
 SUPPLY: VDDA (1, 43, 46), VDDA3P3 (3, 4), VDD3P3_RTC (19), VDD3P3_CPU (37) on +3V3; C82 10 uF + C76 100 nF at VDDA3P3 pins 3/4 (Fig. 11 C1/C2). The 4 MB flash, 40 MHz crystal, CAP1/CAP2 network and the VDD_SDIO capacitor are inside the SIP (Fig. 8); VDD_SDIO (26) stays open.
 EN: CHIP_PU (9) = RX_EN, R56 10k to +3V3 + C73 1 uF (tau 10 ms), never floating.
 STRAPS at reset (P4-6, checked on the netlist):
 GPIO0 (23) RX_BOOT: internal 45k pull-up, D8 blocks -> 1 = SPI boot (SW1 held -> download).
 GPIO2 (22) PA_EN: internal 45k pull-down + LP5907 EN 1 MOhm pull-down (VTX sheet), no external part -> 0.
 GPIO5 (34) VTX_SPI_CLK: internal pull-up, R81 1k into the RTC6705 CMOS input -> 1 with +3V3_VTX up; reads 0 in HD mode (RTC6705 unpowered, its input clamps the pull-up to about 0.6 V), which only changes the SDIO-slave timing (unused).
 GPIO12 / MTDI (18): no connect, internal pull-down -> VDD_SDIO 3.3 V for the in-package flash (VDD_SDIO eFuse burn optional).
 GPIO15 / MTDIO (21): no connect, internal pull-up -> boot log on U0TXD (FC ignores it).
 NOT CONNECTED: flash CMD/IO11 (30) and CLK/IO6 (31); NC pins 25, 35, 36, 44, 45, 47, 48; free IO7/8/9/10 (32, 33, 28, 29; PSRAM on the -02 alternate), IO20 (27), I35 (11), I38 (7), I39 (8).
 ELRS LAYOUT JSON (D53, D81): serial_Lrx 3, serial_Ltx 1, radio_sck 25, radio_mosi 32, radio_miso 33, radio_nss 27, radio_rst 26, radio_busy 36, radio_dio1 37, led_rgb 22, button 0, vtx_nss 19, vtx_sck 5, vtx_mosi 14, vtx_miso -1, vtx_amp_pwm 13, vtx_amp_vpd 4, vtx_amp_vref 2, "vtx_miso": -1 is written explicitly: an overlay only replaces keys, so an omitted key keeps the stock 23 = the PICO flash DI, which SPIClass(HSPI).begin() turns into an input (flash cut, crash at VTX init); -1 makes the core attach the HSPI default GPIO12 (no connect here) instead, patched ELRS: GPIO21 VTX_PWR_CTL (+3V3_VTX LDO EN through R84 1k), GPIO34 PA_NTC (ADC1, read in the iStick window, erratum 3.11).
 VTX LINES: series 1k R80/R81/R82 are drawn on the VTX sheet and placed at the ESP32 end (back-feed limit with +3V3_VTX off, F21). GPIO13 VTX AMP_PWM = ELRS vtx_amp_pwm (RTC6705 PAOUT1 feed PNP, not the SE5004L bias). PA_DET is ADC2_CH0 (not readable with Wi-Fi on; not a flight state).
 +3V3 LOAD: ELRS about 0.07 A in flight, Wi-Fi TX peaks 370 mA (DS Table 19, 802.11b 19.5 dBm) in ELRS Wi-Fi mode only (disarmed, bench); +3V3 peak about 0.48 A with the FC and LED, inside the LP5912 500 mA rating; LDO heat and the COUT sum belong to the POWER sheet (P4-12).
 PLACEMENT: SIP site from floorplan v3 (D55, D61); no 85 C halo rule (D69). Alternate part ESP32-PICO-V3-02 (pins 28/29 unused here, P4-14).

File: rx_esp32_sx1281.kicad_sch

Title: ELRS 2.4 GHz RX: ESP32-PICO-V3 + SX1281

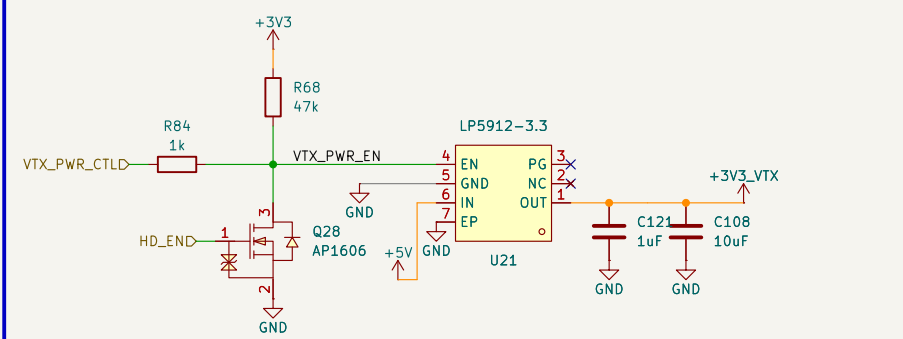
Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

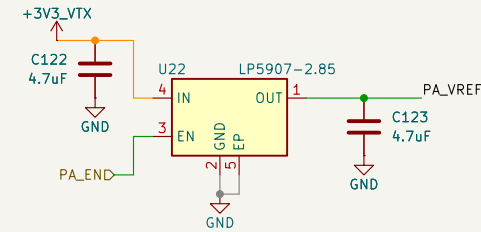
Id: 11/14

+3V3_VTX LDO + HD GATE



U21 LP5912-3.3 on +5V; CIN = the shared 10 uF C13 on POWER (BR-14) only if P5 puts C13 within 1 cm of the U21 IN pin (LP5912 DS 9.2.2.2; tools/check_p5_conditions.py), else U21 gets its own 10 uF 0402 CIN (GHW15SC00106ME110). EN = VTX_PWR_EN; R68 47k to +3V3 (stock ELRS runs the VTX); R84 1k from ESP32 GPIO21 (patched-ELRS power-down hook); GPIO21 low gives VEN 0.07 V < VEN(OFF) 0.3 V; Q28 AP1606 drain = HD gate (HD_EN high: <= 220 mOhm at VGS 2.5 V sinks 70 uA, 3.3 mA with GPIO21 high; VEN < 1 mV). Hot margin (P4 r4, R68 was 100k); EN stays >= VEN(ON) 1.3 V max up to 42 uA of Q28 leakage (100k: 20 uA); AP1606 IDSS <= 1 uA at 25 C, <= 100 uA at 125 C (VDS 20 V), about 16 uA at 85 C; HD_EN before Betaflight drives PIN01; <= 0.22 V vs VGS(th) 0.4 V min at 25 C (250 uA), about 0.25 V at 85 C (-2.5 mV/K); a hot low-Vin part may keep the VTX off until PIN01 is driven low (V9 hot-soak boot check). COUT on +3V3_VTX: C108 10u + C121 1u + C122 4.7u + 3x 100n + 2x 100p = 16.0 uF nominal, about 6.8 uF effective at 3.3 V typical (5.4-8.2 uF at +/-20 %; POWER-sheet DC-bias figures). Inside the LP5912 0.7-10 uF window, C108 sits at the RTC6705 VCO pins 39/40 (VCO bulk). Load 0.10 A typical, about 0.17 A at the top PA step (root power table).

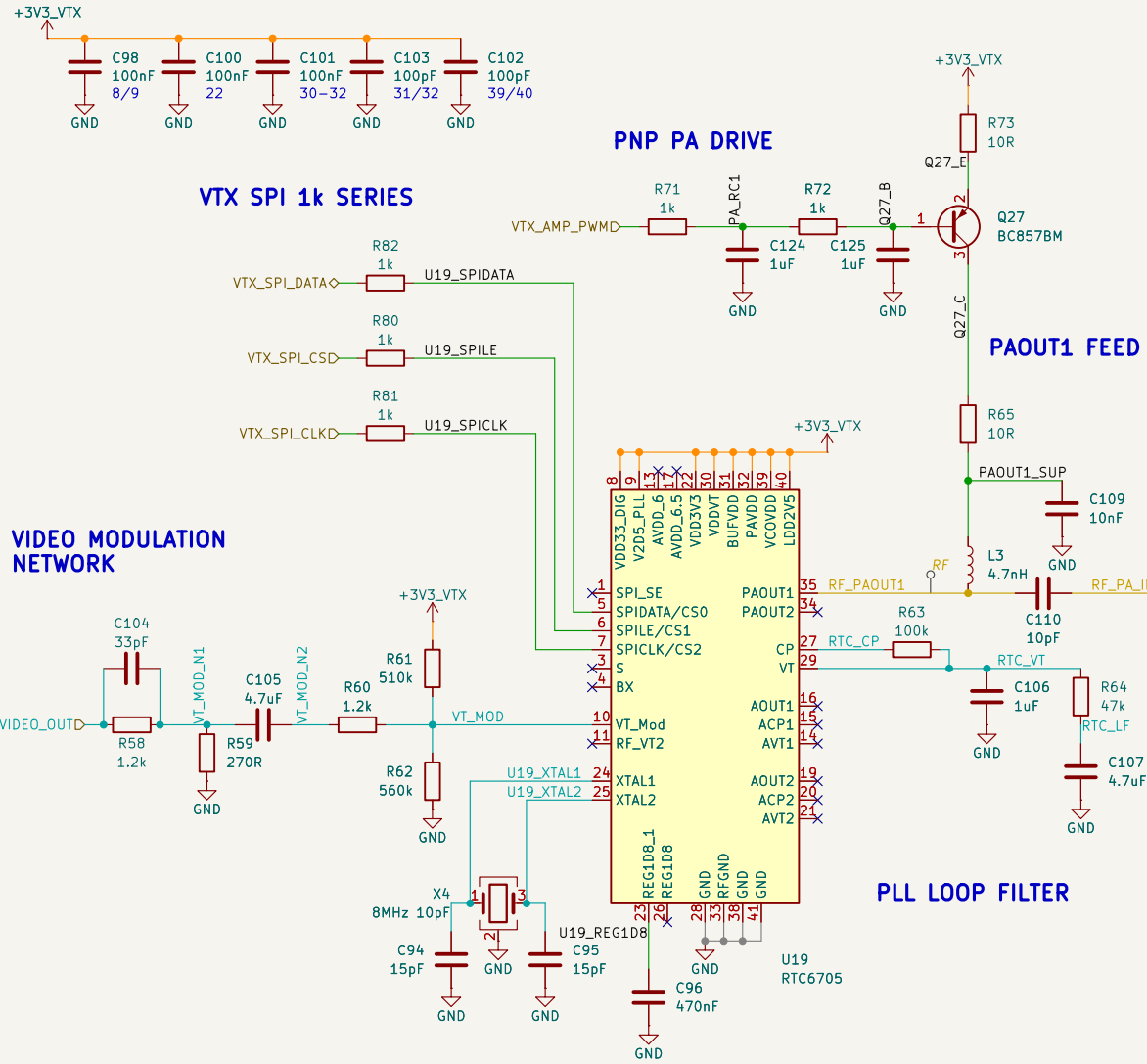
PA REFERENCE LP5907-2.85

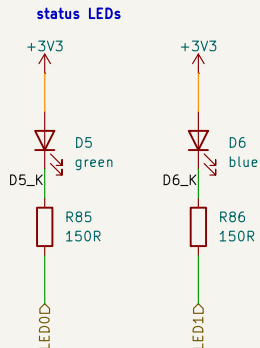


U22 LP5907-2.85 from +3V3_VTX (TI SNV5798); HD mode removes PA_VREF with the rail. EN = PA_EN straight from ESP32 GPIO2 (strap: internal 45k pull-down + the LP5907 1 MOhm EN pull-down). C123 4.7 uF 0402 (> 0.7 uF effective); CIN = C122 4.7 uF 0402 X5R at U22 IN (about 2.6 uF at 3.3 V, >= 2 uF at -20 % and hot); LP5907 DS 5.6 / 7.2/2.4 needs > 0.7 uF effective over all conditions within 1 cm; the 1 uF 0201 (about 0.45 uF) alone missed it (PA critique 3). DEVIATIONS, bench-gated: (a) VREF window: 2.85 V +/- 2 % = 2.79-2.91 V vs SE5004L VREF H 2.80-2.90 V, and the +/- 2 % is specified only for VIN >= VOUT + 1 V (VIN here 3.3 V); V5 measures PA_VREF on several units, cold and hot. (b) HD switch-on with ELRS holding PALEN high: +3V3_VTX collapses before C123 discharges, so VOUT exceeds VIN + 0.3 V (LP5907 abs max, DS 5.1) by up to one body-diode drop (<= 0.7 V, < 1 ms, <= 10 uJ from C123); V9 scopes PA_VREF and +3V3_VTX (< 0.5 V). Fallbacks: C123 down to 1 uF, or the patched ELRS drops PALEN on HD_EN.

RTC6705 VTX: 8 MHz CRYSTAL, SPI, VIDEO, PLL, PA DRIVE, PAOUT1 FEED

Decoupling at the RTC6705 supply pins (pin numbers next to each capacitor):





STATUS LEDs (D5 D6 R85 R86) source: OpenFC-Lite-Mini hardware/rp2350a.kicad_sch 'status LED' block (R52 + R53 2x 75R, D2 XL-1005UBC), copied and adapted: sink drive per hardware/sch_contract.json and research/PINMAP.md, 2x 75R -> one 150R 0201, second column for LED1.
 Circuit: +3V3 -> LED -> 150R -> GPIO, the pin sinks (contract). Betaflight default polarity, low = on, no LEDx_INVERTED: LED0_PIN PA7 (GPIO7, green D5), LED1_PIN PA26 (GPIO26, blue D6).
 Datasheet check (XINGLIGHT XL-1005UGC / UBC, 0402): VF 2.7-3.3 / 2.6-3.4 V at 20 mA, IF max 20 mA, PD 70 mW, VR 5 V. At a few mA the InGaN VF is about 0.2-0.3 V lower (inferred), so with VOL <= 0.5 V at 4 mA drive (RP2350 D5) the LED current is about 0.5-5 mA, typ. 2-3 mA, hard ceiling (3.3 - 2.4) / 150R = 6 mA; LED, 0201 resistor (<= 5.4 mW of 50 mW) and GPIO (4 mA setting, 100 mA IOVSS sum) all inside their ratings; brightness at 1-3 mA is about 17-135 mcd green and 4-36 mcd blue (lv scaled from 350-900 / 80-240 mcd at 20 mA), enough for an indicator. Reverse bias is 0 V when the pin drives high (IOVDD = +3V3).
 Reset state: RP2350 pads come up with the pull-down on (PDE = 1, 36-113 kOhm), so both LEDs may glow faintly (about 10-30 uA) until Betaflight drives the pins and in BOOTSEL mode. Accepted, cosmetic.
 Placement (P5): bottom side, fc_core region (bom_plan); the R may sit at either end of the LED, slow signal (D33). Pins LED0/LED1 = root nets /LED0, /LED1 (RP2350A).
 P4 closed here: bom_plan R85/R86 'P4 checks brightness vs VF' (above). P4-14 fields: RC0201FR-07150RL is a stocked Yageo RC_L part (Element14 4008725, RS 198-7568: 150R 1 % 50 mW 25 V); no LCSC number known, field left empty (contract open item A9-LCSC).

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Sheet: /LED/
 File: led.kicad_sch

Title: Status LEDs

Size: A5 Date: 2026-10-09
 KiCad E.D.A. 10.0.6

Rev: rev1
 Id: 13/14

Battery pads (15)

Source: OpenFC–Lite–Mini hardware/pads.kicad_sch (Conn_01x01 pad, pin down, rail symbol at the wire end).

J2 '+' = +BATT_IN, J3 '-' = GND; BT2.0 pigtail (22 AWG, user–fitted) through plated D1.1 holes in 3.0 x 2.0 mm pads (lib:BattPad_3.0x2.0mm_PTH1.1), >= 12 power vias each into the pours (spec 4.10, D72 power plan: +BATT_IN into L4 + L1/L6, GND into L2/L5). Silk '+' / '-' 2.0 mm and '15' (spec 13).

15 LiPo/LiHV only, 3.0–4.35 V (D14), D1 SMF5.0A TVS, the pad bulk and the shunt R1 sit on POWER, which also holds the +BATT_IN and GND PWR_FLAGS. Netclass VBAT: directive here on +BATT_IN at J2.

Copper only: DNP, out of BOM and CPL (LINEUP A16).

Motor lands M1–M4: plug–ready, Molex PicoBlade header DNP

Source: new (D65: plug–ready lands as on the BetaFPV Matrix 1S 5IN1 II), drawn in the house pad style, One land per motor, lib:MotorPad_PicoBlade_1x03_P1.25mm_PTH0.5: pads 1 / 2 / 3 = Mn_C / Mn_B / Mn_A = /ESCn/PHASE_C / B / A (symbol mirrored so A reads on top; phase order is free, Bluejay sets the direction). All four lands are placed on the top (F; bom_plan side = board), beside the corner ears (J4/J7 at one corner, J10/J13 at the opposite one), mouths facing out (D72).

Motor wires solder to the land (pads 1/3 1.65 x 1.50, pad 2 0.90 x 1.50) or through the 0.50 mm plated holes: the recommended default. The Molex 53047–0310 PicoBlade header (board side of the hobby 'JST1.25' motor plug, wire side 51021–0300) is a DNP BOM line, out of the CPL, hand–fitted from the far side with its pins trimmed on the land side, Molex rates 1.0 A per contact, –40.85 C, far below the motor current (4.3–5.2 A per channel, 12 A bursts); the published ESC ratings apply to soldered motor wires; the README pinout states the 1.0 A contact rating and that plugged motors run outside it at the user's risk (BetaFPV Matrix practice); V4 records the header temperature. Mating zone (D72): the plug faces outward over the ear neck, clear of parts.

Net names come from the ESC sheets (the root wires carry no label: /ESCn/PHASE_x). Netclass Phase: directive on M1_A here, the pattern /ESC?/PHASE_? assigns the rest. Phase copper runs from the AGM210MAP phase island straight into the land; solder areas 0.30 mm and phase copper 0.20 mm from the edge (SOLDER_PAD class). M1 rear right (ESC1), M2 front right (ESC2), M3 rear left (ESC3), M4 front left (ESC4); Betaflight quad–X.

USB: JST SH 1.0 mm 4–pin + ESD

Source: OpenFC–Lite hardware/pads.kicad_sch (JST SH SM04B plug in a frame; there it carries a UART). The USB pinout, D7 and the VBUS flag are new (D57). Commons block USB_C_GCT4105_USBLC6 checked, not used (USB–C and a 6–pin ESD part).

J31 JST BM04B–SRSS–TB(LF)(SN) as in bom_plan (vertical, bottom): 1 GND, 2 D– (USB_D_N), 3 D+ (USB_D_P), 4 VBUS, tabs 5/6 GND = BetaFPV 'USB(GND_DM_DP_5V)' order (bom_plan, IO–SET row 6, LIBRARY D57). Pin 1 = GND is inferred: V–plan gate V0–USB before the P5 freeze (continuity on a BetaFPV SH1.0 USB adapter or a Matrix II close–up); a mirrored plug would put 5 V on board GND and –5 V on the SGM40661 IN (abs min –0.3 V), D76 (owner) keeps the vertical BM04B facing up next to a mounting hole (D72 side–entry SM04B withdrawn); its mating zone above the board stays clear of tall parts.

D7 TPD2EUSB30DRTR (TI SLVSAC2G): pin 1 D+, 2 D–, 3 GND (Table 5–1); VWRM 5.5 V, IO abs max 6 V, so a D+/D– short to VBUS survives; 8 V clamp at 1 A, 0.7 pF, IEC 61000–4–2 8 kV contact, –40.85 C. It sits at J31, ahead of the 27 R R42/R43 at the RP2354A (RP2350A sheet). Layout per DS 10.1: next to J31, protected traces straight with rounded corners, no unprotected trace beside the J31–D7 stretch, GND pin by via–in–pad into L2. USB pair 90 ohm, length–matched (D33); netclass USB from the directive on USB_D_P (_P/_N pair).

VBUS is a hierarchical net (LINEUP A11) to POWER: U24 SGM40661 OVP IN + C131 100 nF take the hot–plug and over–voltage duty (D57). PWR_FLAG here: J31 is the only source of VBUS. A cable with pins 1/4 swapped puts –5 V on VBUS (unprotected, as on the Matrix).

LED strip + buzzer row

Source: OpenFC–Lite–Mini hardware/pads.kicad_sch (LED–strip and buzzer pads).

Row GND LED 5V BZ– (D63/D64): the strip gets its own GND pad J33 (supersedes D27).

LED = LED_STRIP driven straight from GPIO8 (PIO1 ws2812; D52 BR–04; no translator, no series resistor, Matrix II parity). A WS2812B on 5 V asks VIH 0.7 VDD = 3.5 V; the 3.3 V drive works on the common whoop strips, a level problem shows as a dead first LED (bench check, V9).

5V = +5V for the strip (0.20 A allowance, analog mode, spec 5.1) and the buzzer +, BZ– = BUZZER–: drain of the AP1606 low–side switch Q26, gate on GPIO17 (BEEPER_INVERTED, RP2350A sheet); the buzzer sits between 5V and BZ–.

Copper only (DNP, out of BOM and CPL, LINEUP A16).

Camera pads + JST SUR 0.8 mm plug

Source: OpenFC–Lite hardware/pads.kicad_sch (CAM frame: pads + SH1.0 SM03B plug), adapted to the JST SUR 0.8 mm SM03B–SURS–TF(LF)(SN) (D63/D64, IO–SET row 3: BetaFPV C03 and the Matrix–recommended cameras).

Plug and pads in parallel: J32 pin 1 VIDEO_IN, 2 GND, 3 +5V_CAM, tabs 4/5 GND: pads CAM / GND / 5V (J23 J25 J24). On the board the plug faces inward and the pads sit in front of its mouth: plug OR solder in one mating zone (D72).

VIDEO_IN –> OSD front end (TLV7031 sync + SN74LVC1G3157, OSD sheet), netclass Analog (OSD directive), +5V_CAM = +5V after FB1 120 R + C15 10 uF (POWER), 0.12 A. Video return: CAM GND pad and plug pin 2 enter L2 at the CAM pad (spec 11).

Gate G1 still open: pin order provisional until calipers + photo on a C03 lead and the MX2 CAM IN. With GND in the middle a mirrored footprint puts 5 V on the camera's video output. G1 = 1.0 mm pitch –> SM03B–SRSS–TB(LF)(SN), same pins.

HD pads (DJI O4 Lite)

Source: OpenFC–Lite–Mini hardware/pads.kicad_sch (digital video pads), new pad set (D13/D16).

VHD = +5V_HD from U5 TPS22810 (on with HD_EN and cell >= 3.23–3.82 V, held down to 2.74–3.14 V; POWER P4–1, 1 % resistors), 1.20 A budget; O4 Lite input 3.7–13.2 V, TX1 = UART1_TX (GPIO4) –> O4 RX, RX1 = UART1_RX (GPIO5) <– O4 TX; MSP DisplayPort; no SBUS pad (D16). Cut DJI cable: red –> VHD, black + brown –> GND, white –> TX1, grey –> RX1, yellow open (no SBUS input since D89).

Analog builds: TX1/RX1 are the free hardware UART (the only one since D89, no user UART pads), or I2C0 via the I2C–HD preset (IO–SET row 8).

P4–11: the O4 stays unpowered until HD_EN, so its TX cannot drive RX1 before IOVDD (FT pads limited to 3.63 V unpowered, PINMAP F16). A device on TX1/RX1 in an analog build runs from a 5V pad (LED row), which comes up before +3V3; 3.3 V logic only (a 5 V TX on RX1 can exceed 3.63 V during the ramp).

Copper only (DNP, out of BOM and CPL, LINEUP A16).

Mounting holes

H1–H3: the 25.5 mm M1.4 pattern (D73) is cut in Edge.Cuts (ear holes with flange keep–outs, spec 7), so the symbols are excluded from the board (FLOORPLAN 'For P4'). BOM and CPL: footprint kept as reference. No net (non–plated; GND annulus = O16). LOG01 (incutec, B.Silkscreen) is a board–only footprint, no symbol.

PADS: every off–board connection. Style from the OpenFC–Lite–Mini and OpenFC–Lite pads sheets (pads pin down, a rail symbol or a hierarchical label at the wire end, plug blocks in frames); the source of each block is in its note. References, values (pad codes = silk codes), footprints, MPN / Manufacturer / LCSC and the DNP / BOM / CPL attributes are hardware/bom_plan.json; solder pads are DNP and out of BOM and CPL (LINEUP A16), the motor header is a DNP BOM line, 20 hierarchical labels = the PADS sheet pins (sch_contract.json): VIDEO_IN, M1_A..M4_C, UART1_TX/RX, LED_STRIP, BUZZER–, USB_D_P/N, VBUS, Rails by power symbol (+BATT_IN, +5V, +5V_CAM, +5V_HD, GND; sources on POWER); VBUS carries its PWR_FLAG here. Netclass directives: VBAT (+BATT_IN at J2), Phase (M1_A), USB (USB_D_P). No user UART group (D89): the free UART is the HD group TX1/RX1 (UART1).

Owed: J31 pin 1 = GND (gate V0–USB before the P5 freeze), J32 pin order (gate G1), motor–header temperature (V4).

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Sheet: /PADS/ File: pads.kicad_sch		
Title: Pads, motor lands, camera, HD, USB		
Size: A3	Date: 2026–10–09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 14/14